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(12) **United States Patent**
Stankey et al.(10) **Patent No.:** **US 12,404,524 B2**(45) **Date of Patent:** ***Sep. 2, 2025**(54) **EXPRESSION VECTOR**(71) Applicant: **TERRA BIOWORKS, INC.**,
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Middleton, WI (US)(*) Notice: Subject to any disclaimer, the term of this
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U.S.C. 154(b) by 96 days.This patent is subject to a terminal dis-
claimer.(21) Appl. No.: **17/813,088**(22) Filed: **Jul. 18, 2022**(65) **Prior Publication Data**

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Related U.S. Application Data(63) Continuation of application No. 16/815,399, filed on
Mar. 11, 2020, now Pat. No. 11,390,882.(60) Provisional application No. 62/817,345, filed on Mar.
12, 2019.(51) **Int. Cl.****C12N 15/52** (2006.01)**C12N 9/00** (2006.01)**C12N 15/09** (2006.01)**C12N 15/70** (2006.01)**C12N 15/86** (2006.01)**C12Q 1/68** (2018.01)**C40B 40/08** (2006.01)(52) **U.S. Cl.**CPC **C12N 15/86** (2013.01); **C12N 15/52**
(2013.01); **C12N 15/70** (2013.01)(58) **Field of Classification Search**CPC **C12N 15/86**; **C12N 15/52**; **C12N 15/70**;
C40B 40/08

See application file for complete search history.

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ABSTRACT

Disclosed herein are recombinant methods of activating expression of one or more biosynthetic gene clusters comprising more than one gene, the method comprising a recombinant DNA expression vector that possess two opposable inducible promoters that drives expression of a biosynthetic gene cluster exogenously from outside of the cluster to produce polyketides or non-ribosomal peptides in a heterologous host.

16 Claims, 9 Drawing Sheets**Specification includes a Sequence Listing.**

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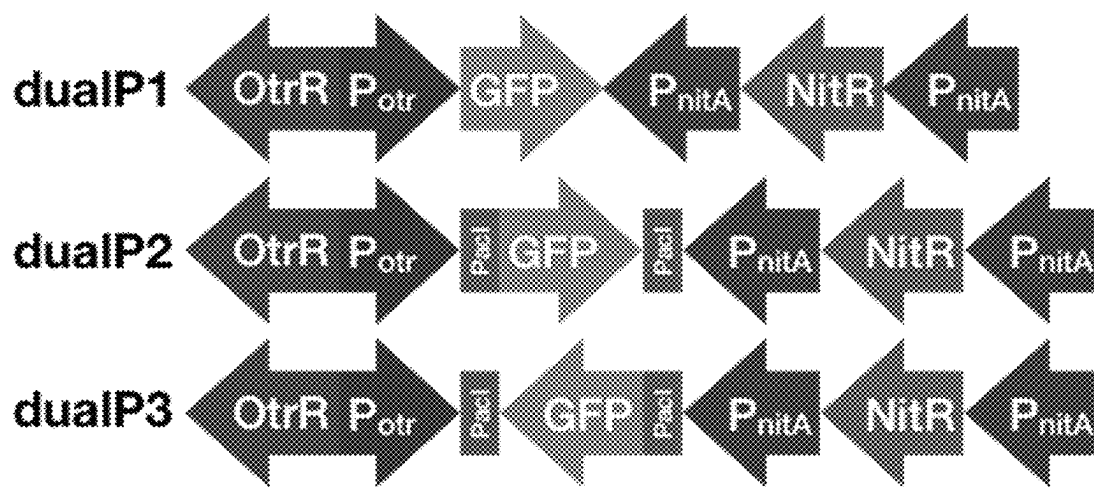


FIG. 1

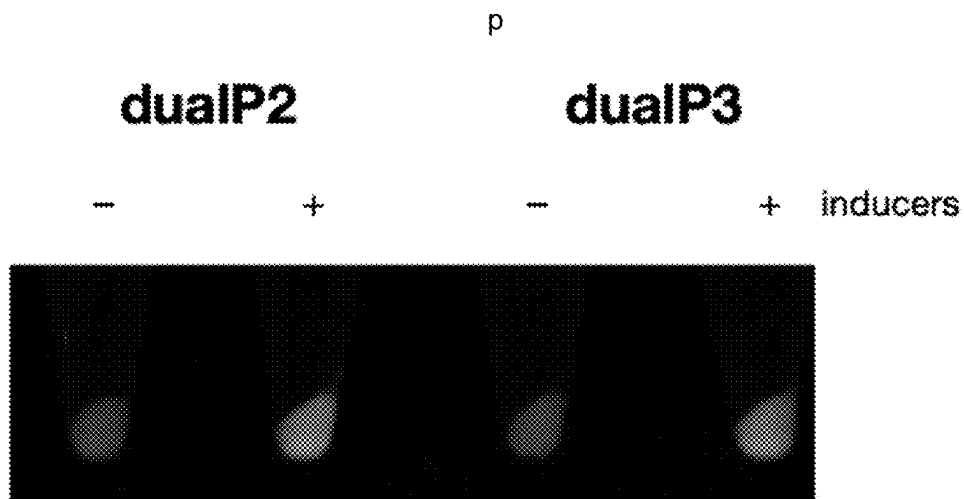


FIG. 2

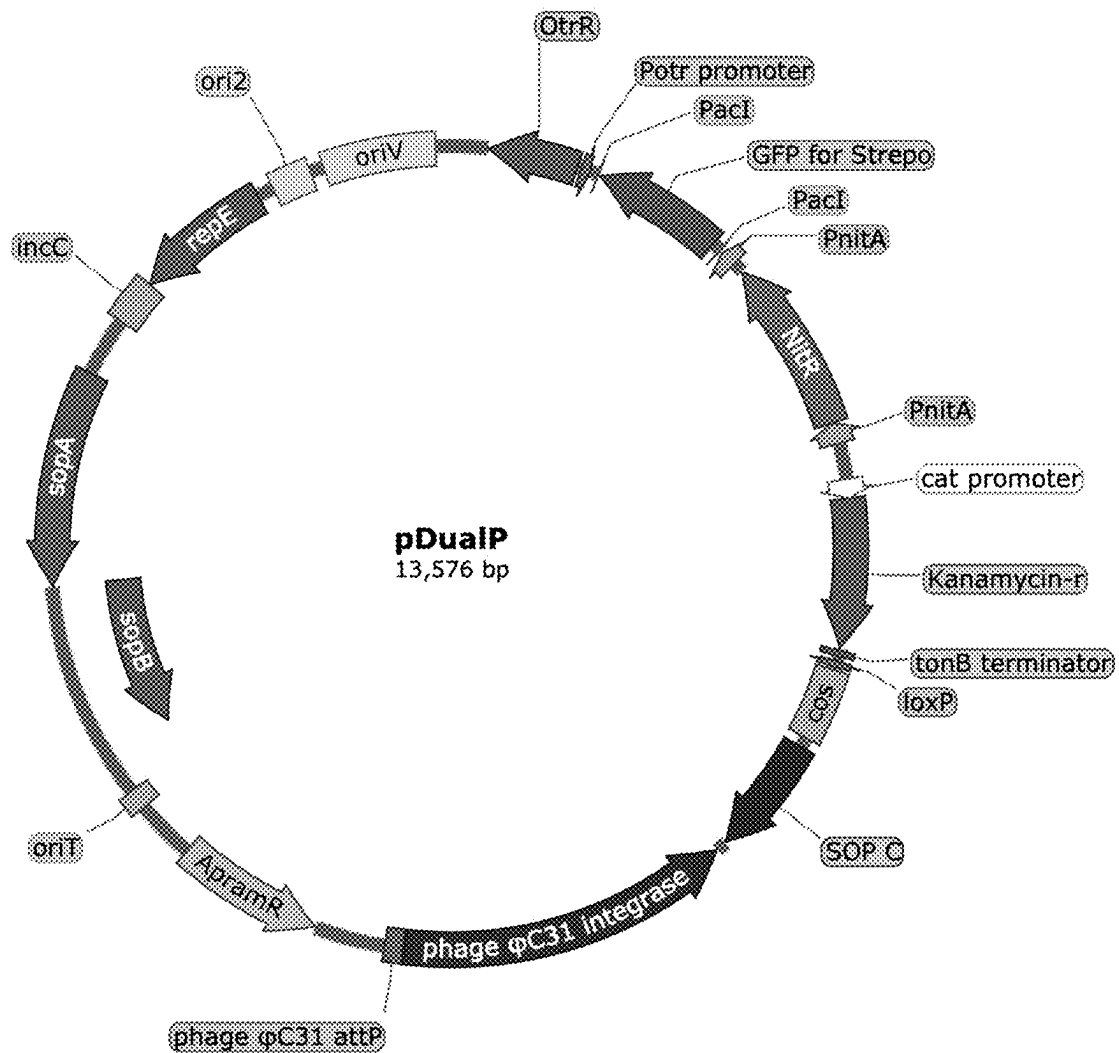


FIG. 3

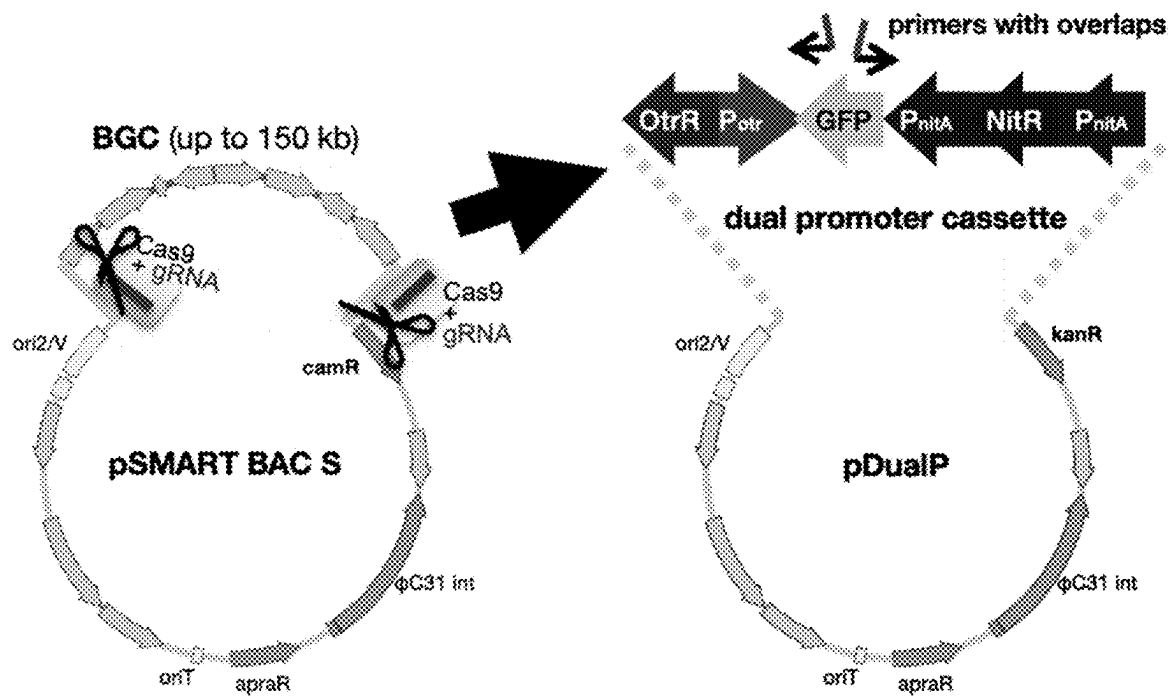


FIG. 4

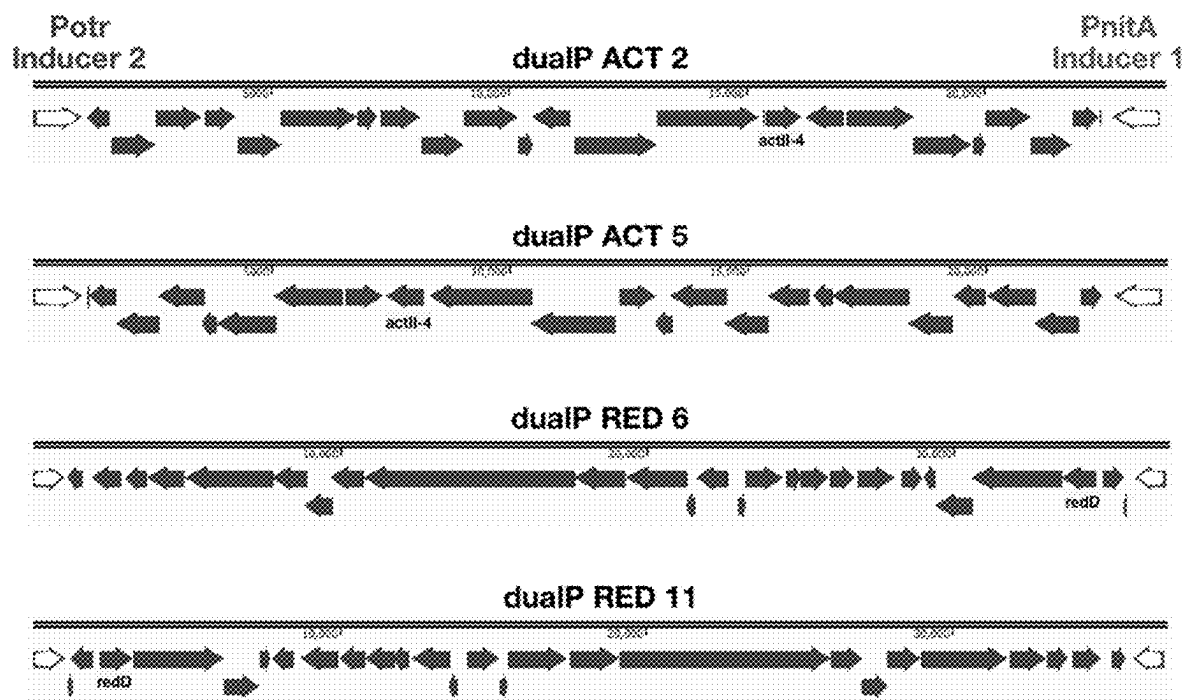
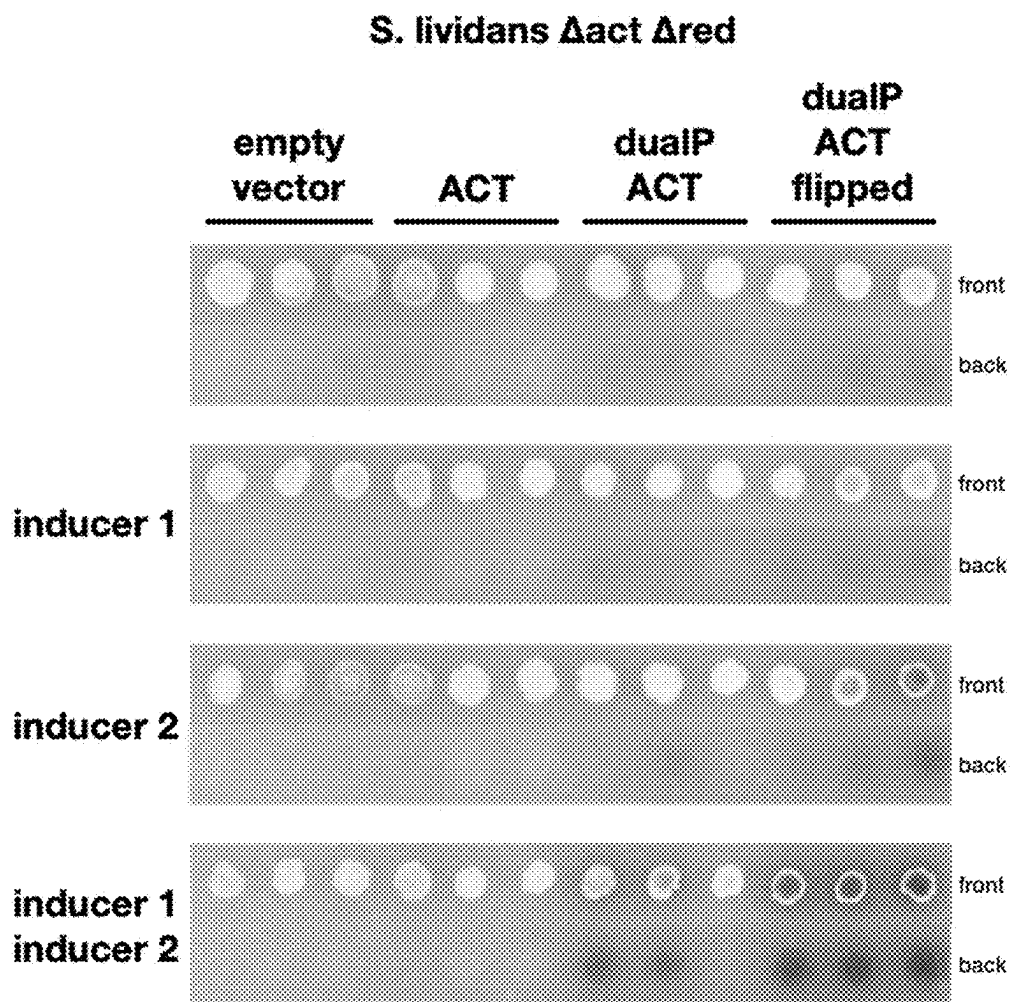
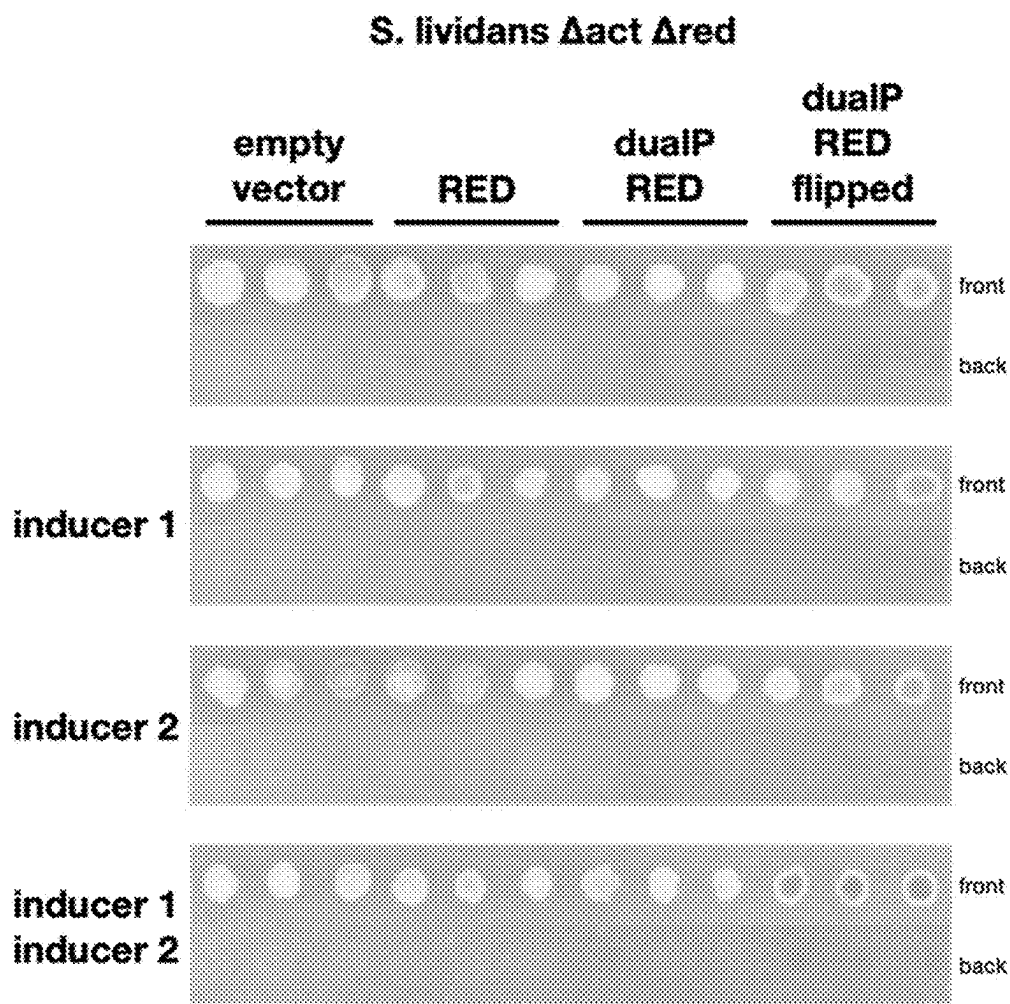


FIG. 5



Spot 1 μ L of YEME liquid culture to MS agar
Imaged after 5 days at 25 °C

FIG. 6



Spot 1 μ L of YEME liquid culture to MS agar
Imaged after 5 days at 25 °C

FIG. 7

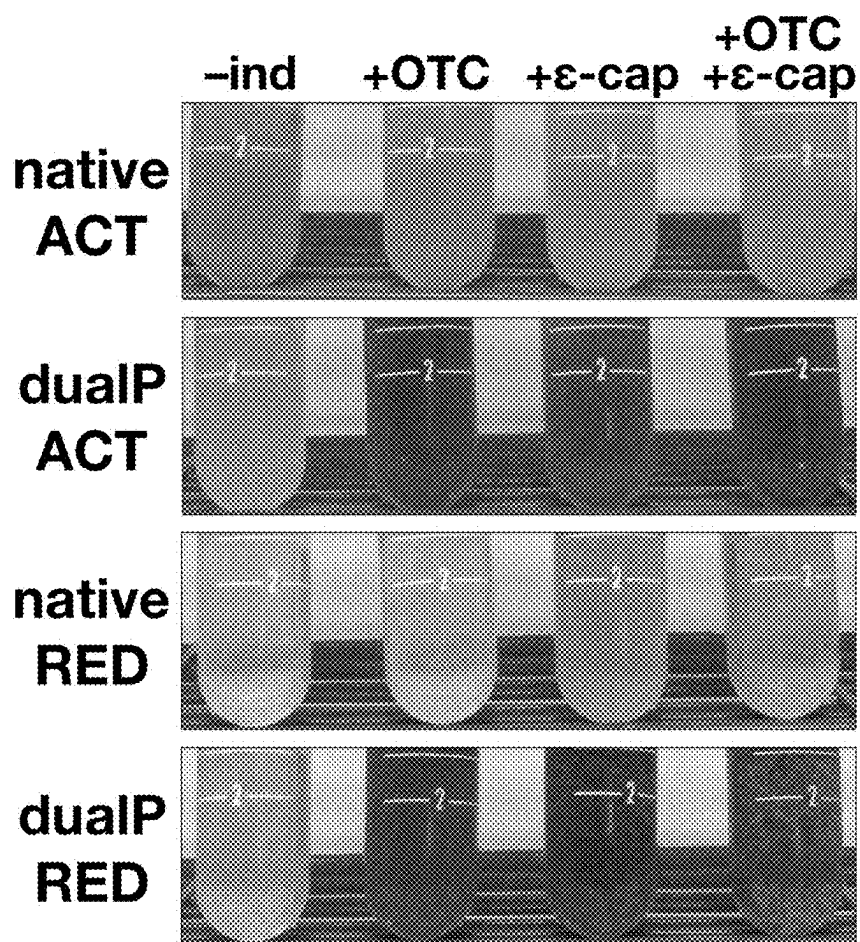


FIG. 8

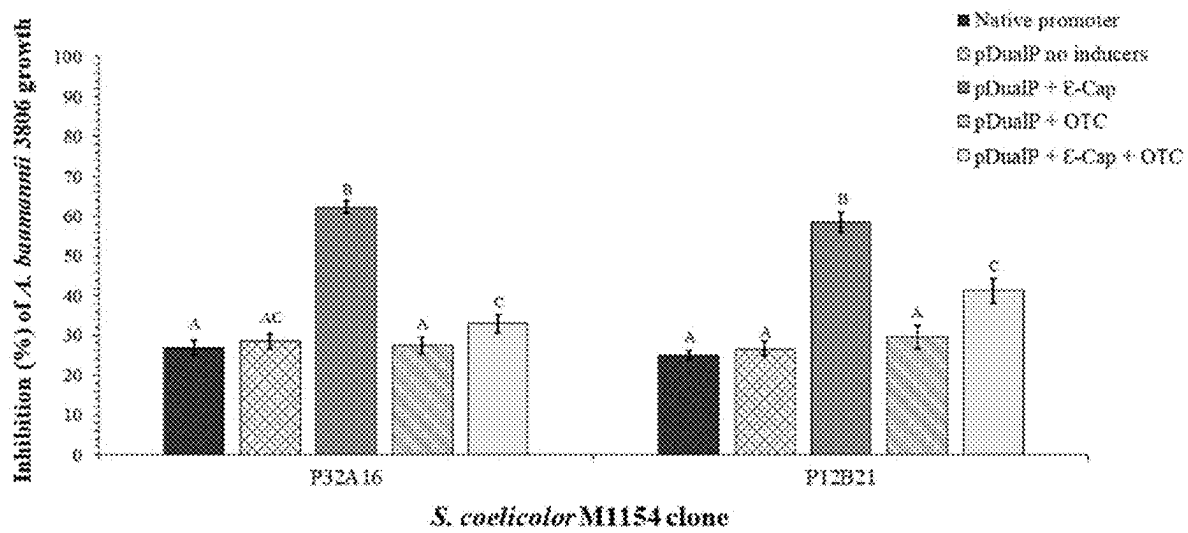


FIG. 9

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EXPRESSION VECTOR

This application is a continuation of Ser. No. 16/815,399, filed Mar. 11, 2022, which claims priority to U.S. provisional patent application Ser. No. 62/817,345, filed Mar. 12, 2019, which is incorporated herein by reference in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

This invention was made with government support under grant numbers 5R44AI085840-04 and R43AT008295 awarded by the National Institutes of Health. The United States government has certain rights to the invention.

SEQUENCE LISTING

The text of the computer readable sequence listing filed herewith, titled "VARI-37579-303_SQL", created Jul. 18, 2022, having a file size of 134,699 bytes, is hereby incorporated by reference in its entirety.

FIELD

Provided herein is technology related to expression and discovery of natural products and biologically active agents and particularly, but not exclusively, to compositions, methods, systems, and materials for expressing biologically active agents by recombinant DNA technology. The technology finds use in, e.g., the fields of agriculture, chemistry, medicinal chemistry, medicine, molecular biology, and pharmacology.

BACKGROUND

Small molecule compounds produced by bacteria, fungi, and plants hold tremendous potential for new pharmaceuticals, therapeutic agents, and industrially useful compounds. Over 60% of clinically useful anticancer drugs and 49% of anti-infective drugs in use today are derived from secondary metabolites called natural products (NP). Despite this record, pharmaceutical companies have cultivated millions of microbes searching for new bioactive natural product compounds only to rediscover known chemical scaffolds >99% of the time. In contrast, whole genome sequencing and metagenomic approaches (e.g., studies of DNA isolated from complex microbial communities) reveal an immense diversity of genes encoding unknown metabolites that are missed by conventional cultivation-based screening approaches. Culture-independent studies (e.g., metagenomics) reveal that only a small fraction of NP pathways have been expressed to produce a known metabolite. Accessing these compounds promises to reinvigorate drug discovery pipelines and provide novel routes to synthesize complex chemicals.

Polyketides, ribosomally synthesized and post-translationally modified peptides, and nonribosomal peptides are important classes of natural products responsible for the development of many human therapeutic, veterinary, and agricultural products (e.g., FK506, lovastatin, avermectin, vancomycin, daptomycin, and teixobactin). Genes encoding NP are usually arranged in contiguous units called biosynthetic gene clusters (BGCs) in bacteria and fungi, making it possible to identify their contents, e.g., using bioinformatic analysis tools (e.g., antiSMASH (Weber et al. 2015), BAGEL3 (van Heel A J, de Jong A, Montalban-Lopez M et

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al. BAGEL3: Automated identification of genes encoding bacteriocins and (non-) bactericidal posttranslationally modified peptides. *Nucleic Acids Res* 2013; 41: W448-53, incorporated herein by reference), and PRISM (Skinnider M A, Johnston C W, Edgar R E et al. Genomic charting of ribosomally synthesized natural product chemical space facilitates targeted mining. *P Natl Acad Sci USA* 2016; 113: E6343-51, incorporated herein by reference)) to recognize structural motifs found within the encoded enzymes. Some enzymes that synthesize these compounds—e.g., polyketide synthases (PKS) or nonribosomal peptide synthases (NRPS)—typically exist as large multi-modular scaffolds that have been the target of various molecular engineering methods directed to producing more of the small molecule compounds or improved analogs of existing pharmaceuticals. Further complicating things, many NP biosynthetic pathways comprise dozens of genes and are often 5 to 200 kilobases (kb) in size; thus, special tools and technologies are often used for biosynthesis of NPs.

A major bottleneck in NP discovery is the production of sufficient metabolite for biochemical, structural, and/or cytotoxicity analyses. Classical strain improvement and process development programs sometimes involve years of work to increase the yield of a compound from the natural producing organism and have achieved greater than 100-fold increases in titers. A number of microorganisms have been optimized through random mutagenesis for bulk production of highly valuable compounds, including penicillins, macrolide antibiotics, and lovastatins. However, this conventional approach to strain improvement is not feasible during the early stages of discovering or characterizing NPs.

Ribosome and RNA polymerase engineering, regulatory gene activation, the use of eliciting agents, epigenetic perturbations, testing multiple media recipes, and many of the other recent methods developed to improve NP expression are low throughput and are limited to being applied to one pathway or one organism at a time. Refactoring a NP pathway in a native host organism to insert promoters or modify regulatory elements is a relatively new method to overproduce metabolites that involves the use of specialized genome editing tools, a genetically tractable host, and months of effort to identify the proper combination of genetic elements that increase metabolite production. An alternative method comprises cloning the pathway onto a shuttle vector and moving the cloned pathway into a surrogate "heterologous host" species that is often engineered to enhance the production and discovery of secondary metabolites. This process does not always produce detectable metabolites or sufficient metabolite to characterize the molecule further. Accordingly, such approaches often involve additional efforts to modify the cloned pathway to overexpress the proteins needed to produce the small molecule.

The development of new drugs from NP biosynthetic pathways is labor intensive and very expensive. Current methods involve specific and particularized tailoring and precise genetic modifications for each pathway, most of which involve multiple gene products encoded by very large gene clusters that are difficult to manipulate. Nevertheless, heterologous expression of entire BGCs in a genetically tractable host is one of the most promising approaches to connect BGCs to their NP. Heterologous expression permits the characterization of BGCs from cultured microbes and from metagenomic DNA and provides a technology for accessing potentially new and valuable compounds. However, BGCs often are 20-200 kb in size and their manipulation involves use of specialized cloning methods and autonomously replicating bacterial chromosome vectors

called bacterial artificial chromosomes (BACs) or cosmid/fosmid cloning vectors. Cloning methods often entail producing a clone library of large DNA inserts (e.g., comprising 5 to 10 to 100 kb or more) from a genome or metagenome and then screening the clone library by PCR or other sequence-specific methods to locate a clone comprising the desired BGC. While such approaches have proven to be successful, constructing and managing such a clone library and then finding an entire pathway on one clone is challenging. Some recombination-based technologies have been developed to address library construction by circumventing the entire genome library construction approach of obtaining a BGC. However, these technologies are also lengthy and complicated. For example, recent strategies using CRISPR/Cas9 as a universal restriction endonuclease have been used in vitro to excise a BGC precisely for further manipulation. Such approaches have led to rapid improvements in directly cloning BGCs from essentially any bacterial or fungal genome that has been sequenced.

Heterologous expression of biosynthetic gene clusters in a genetically tractable host can provide a more directed strategy for natural product discovery and a variety of new tools have been developed recently to investigate the vast number of BGCs identified by sequencing microbes. The heterologous expression process usually comprises five steps: assembling a high quality sequence of a target genome or metagenome, identifying a target BGC, cloning the BGC or genes of a BGC that provide a biosynthetic pathway, expressing the genes of the pathway in a heterologous host, and detecting the metabolite produced by the host. Briefly, in some approaches, BGCs are identified from sequenced genomes or metagenomic clones using computational tools such as AntiSMASH. Various DNA cloning and/or assembly tools and engineered heterologous hosts are then used for expression of the large biosynthetic gene clusters. To identify the target natural products, the resulting metabolite profiles are evaluated and characterized by advanced metabolomics and detection techniques such as mass spectrometry and antibiosis activity against pathogenic microbes.

Despite improvements in molecular biological tools, significant issues remain for heterologous expression technologies. For instance, moving a BGC between organisms, even closely related ones, can drastically change the productivity of a pathway because regulatory gene pathways are not fully conserved between species. Accordingly, it is not surprising that most natural product biosynthetic gene clusters identified in microbial genomic and metagenomic sequencing efforts are silent under laboratory growth conditions. It is therefore often necessary to refactor the regulation of a BGC using well-characterized promoters to provide expression in a heterologous host. BGC refactoring decouples gene clusters from native regulatory contexts by placing pathway genes encoded by the BGC downstream of known, characterized promoters in a production host. However, the field insufficiently understands the transcriptional regulation hierarchy to predict precisely the promoter refactoring events that induce secondary metabolite production from most silent biosynthetic gene clusters. Due to this uncertainty, each promoter of a small library of known, different promoters is placed in multiple strategic positions in the pathway in an attempt to identify a combination that activates the cluster, if any. Often times several hundred different combinations of promoters and insertion sites are constructed and screened to produce a compound. This process is lengthy and challenging. The amount of work and time involved in refactoring a single pathway is immense, and

current tools that allow for controllable expression of BGCs do not scale to screening tens or hundreds of novel pathways.

In spite of advances in genome sequencing, bioinformatic discovery of BGCs, and cloning of BGCs into shuttle vectors for manipulation and experimentation, there is still no universal method to activate the expression of large NP pathways. Often the pathway does not produce sufficient or detectable metabolite in its surrogate heterologous host and additional extensive genetic manipulation is required, which may entail many months of effort by multi-functional teams. The current state of the art for expressing BGCs in *Streptomyces* involves genetic engineering of native or synthetic promoters located endogenously within a BGC to attempt to activate the entire pathway. Accordingly, a universal tool that activates quiescent pathways exogenously and/or facilitates over-expression of BGCs would have a major impact in accelerating NP drug discovery by giving researchers more control over expression pathways of interest in ways that were only previously achievable by laborious and time-consuming genetic refactoring.

It is well known in the art that individual genes can be expressed using constitutive or inducible promoters in nearly any host. For example, the inducible promoters Potr (1) and PnitA (2, 3) have been described for use in single-gene expression experiments and their respective oxytetracycline (OTC) and ϵ -caprolactam (ϵ -cap) inducers appear to have no effect on the physiology or growth rate of common *Streptomyces* expression strains. There are numerous examples of promoters being placed strategically in front of one or a few genes in a BGC to initiate expression of some of the proteins in the pathway (see, e.g., Int'l Pat. Pub. No. WO2017151059A1, incorporated herein by reference).

It is also well known in the art that dual promoter vectors have been used to transcribe DNA from one or both sides of a recombinant clone (see, e.g., U.S. Pat. Nos. 4,766,072 and 5,017,488, each of which is incorporated herein by reference) or to express one (see, e.g., U.S. Pat. No. 6,117,651, incorporated herein by reference) or several protein-encoding genes (see, e.g., U.S. Pat. No. 9,546,202, incorporated herein by reference) operably linked to said promoters. There are also multiple examples of uni-directional or bi-directional promoters used to express several genes in a BGC when inserted into the appropriate position between genes within a pathway (see, e.g., Int'l Pat. Pub. No. WO2017151059A1, incorporated herein by reference).

Accordingly, the art would benefit from technology that improves the efficiency, simplicity, and/or throughput of expressing natural products from BGCs and/or from large inserts (e.g., comprising 5 to 10 to 100 kb or more).

SUMMARY

Accordingly, provided herein is a technology in which transcription of a BGC is placed under control of promoters that are present in a cloning vector and that are thus external to the BGC. There are no known examples of a promoter being used outside of the confines of a BGC, that is, placed exogenously outside of the BGC to activate expression of the entire pathway. The present technology teaches for the first time that non-strategically placed exogenous promoters can be used to activate endogenous regulatory components within a BGC that activate the production of a biologically active agent. In contrast to previous technologies, embodiments of the technology provided herein comprise cloning a BGC into an inducible dual-promoter vector to produce a small molecule metabolite. The dual-promoter vector pro-

vides transcription of pathway and/or pathway components (e.g., genes) from one or both directions. In some embodiments, the technology provided herein is a plug-and-play approach for heterologous expression that reduces the amount of time needed to produce a NP metabolite, e.g., from months to days.

Biosynthetic gene clusters (BGCs) encode multiple genes that produce small molecule compounds of considerable therapeutic value as drugs for fighting cancer, viral and bacterial infections, and more. BGCs often contain dozens of genes clustered in a functional unit and comprise multiple promoters and regulator elements in multiple orientations. Activating a BGC to discover novel small molecules or to over-express a BGC to produce more of a known compound, is technically difficult and very complicated.

Accordingly, the technology provided herein facilitates chemical analyses of the biosynthetic potential of gene clusters from genomic or metagenomic sources by improving heterologous expression of BGCs. In particular, embodiments of the technology provide a dual-promoter BAC vector that comprises two inducible promoters flanking a cloning site for heterologous expression of cloned inserts (e.g., comprising a BGC) in *Streptomyces* species. After cloning a BGC nucleic acid into the dual-promoter vector (e.g., in an *E. coli* host) and the presence of the cloned BGC sequence is verified, the recombinant clone is transconjugated to a *Streptomyces* spp. for heterologous expression. The transconjugant strain may now be capable of expressing the BGC using, e.g., basal expression (no inducer) from native promoters or BGC expression may be inducible using one or both of the non-native promoters of the vector that flank the cloned BGC nucleic acid.

During the development of the technology provided herein, experiments using the 21-kb ACT cluster and the 33-kb RED cluster from *Streptomyces coelicolor* (encoding a blue or red anti-microbial product, respectively) indicated that the technology expressed products of the cloned BGC inserts without the vector promoters being operably linked to any gene. In particular, after cloning each of these BGCs in both orientations into the dual-promoter BAC vector and transconjugating the recombinant clones to *Streptomyces lividans* Δ act Δ red (comprising deletions of the chromosomal ACT and RED clusters), the expected blue or red products were produced in an inducible manner. In control experiments, the wild type promoters were not able to express either RED or ACT robustly or in an inducible manner in the heterologous host. During the development of embodiments of the technology, experiments also indicated that the technology also expressed two other BGCs, one encoding a Type I PKS and one encoding a NRPS cluster, that were discovered from a soil metagenomic library. When expressed in *S. coelicolor* M1154 from native promoters, these BGCs each weakly expressed an antibacterial metabolite(s) that inhibited the growth of multidrug-resistant bacterial pathogens (e.g., *Acinetobacter baumannii*). After transfer of these BGCs to the dual-promoter BAC vector and transconjugating to *S. coelicolor* M1154, a significant 2-3-fold increase in antibacterial activity against *A. baumannii* was detected when an inducible promoter in the vector was used to drive transcription relative to antibiotic activity due to expression of the BGCs from native promoters.

Accordingly, in some embodiments the technology provides an expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein the expression vector is configured to accept a biosynthetic gene

cluster nucleic acid at the cloning site and express a product of the biosynthetic gene cluster nucleic acid under control of the first promoter and/or the second promoter. In some embodiments, the first promoter and/or the second promoter is an inducible promoter. In some embodiments, the first promoter and/or the second promoter directs transcription in a host cell that is different than the source of the biosynthetic gene cluster nucleic acid. In some embodiments, the first promoter and/or the second promoter directs transcription in *Streptomyces*. In some embodiments, the first promoter is Potr or Potr*. In some embodiments, the second promoter is PnitA. In some embodiments, the first promoter comprises a nucleotide sequence provided by SEQ ID NO: 1. In some embodiments, the first promoter comprises a nucleotide sequence provided by SEQ ID NO: 2. In some embodiments, the second promoter comprises a nucleotide sequence provided by SEQ ID NO: 6. In some embodiments, the expression vector further comprises OtrR. In some embodiments, the expression vector further comprises NitR. In some embodiments, the vector further comprises a nucleotide sequence provided by SEQ ID NO: 3. In some embodiments, the expression vector further comprises a nucleotide sequence provided by SEQ ID NO: 5.

In some embodiments, the cloning site comprises a restriction enzyme recognition site. In some embodiments, cloning site comprises a restriction enzyme recognition sequence comprising 6, 7, 8 or more nucleotides. In some embodiments, the cloning site comprises an integration sequence adapted to facilitate integration of a nucleic acid by recombination. In some embodiments, the cloning site comprises a multiple cloning site. In some embodiments, the cloning site comprises a selectable and/or screenable marker. In some embodiments, the expression vector further comprises a selectable marker for *Streptomyces*. In some embodiments, the expression vector further comprises a selectable marker for *E. coli*. In some embodiments, the expression vector further comprises an *E. coli* origin of replication. In some embodiments, the expression vector further comprises a *Streptomyces* origin of replication. In some embodiments, the expression vector further comprises a gene that stabilizes large plasmids. In some embodiments, the expression vector further comprises a sopA gene, a sopB gene, and/or a sopC gene. In some embodiments, the expression vector is configured to accept an insert comprising more than 10 kb, more than 20 kb, more than 50 kb, and/or more than 100 kb (e.g., at least 10 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the expression vector is configured to accept an insert comprising 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)).

In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid when the expression vector is present in a host cell. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid when the expression vector is present in a *Streptomyces* host cell. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid that is a nucleic acid or a polypeptide. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid that is a product of one or more enzymes encoded by the biosynthetic gene cluster. In some embodiments, the expres-

sion vector is configured to express a product that is a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomatic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a product of a polyketide synthase or nonribosomal peptide synthase.

In some embodiments, the technology provides a vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein the expression vector is configured to accept a nucleic acid comprising at least 10 kb ((e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)) at the cloning site and express a product of the nucleic acid under control of the first promoter and/or the second promoter. In some embodiments, the expression vector is configured to accept an insert comprising 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the first promoter and/or the second promoter is an inducible promoter. In some embodiments, the first promoter and/or the second promoter directs transcription in a host cell that is different than the source of the nucleic acid. In some embodiments, the first promoter and/or the second promoter directs transcription in *Streptomyces*. In some embodiments, the first promoter is Potr or Potr*. In some embodiments, the second promoter is PnitA. In some embodiments, the first promoter comprises a nucleotide sequence provided by SEQ ID NO: 1. In some embodiments, the first promoter comprises a nucleotide sequence provided by SEQ ID NO: 2. In some embodiments, the second promoter comprises a nucleotide sequence provided by SEQ ID NO: 6. In some embodiments, the vector further comprises OtrR. In some embodiments, the vector further comprises NitR. In some embodiments, the vector further comprises a nucleotide sequence provided by SEQ ID NO: 3. In some embodiments, the vector further comprises a nucleotide sequence provided by SEQ ID NO: 5. In some embodiments, the cloning site comprises a restriction enzyme recognition site. In some embodiments, the cloning site comprises a restriction enzyme recognition sequence comprising 6, 7, 8 or more nucleotides. In some embodiments, the cloning site comprises an integration sequence adapted to facilitate integration of a nucleic acid by recombination. In some embodiments, the cloning site comprises a multiple cloning site. In some embodiments, the cloning site comprises a selectable and/or screenable marker. In some embodiments, the expression vector further comprises a selectable marker for *Streptomyces*. In some embodiments, the expression vector further comprises a selectable marker for *E. coli*. In some embodiments, the expression vector further comprises an *E. coli* origin of replication. In some embodiments, the expression vector further comprises a *Streptomyces* origin of replication. In some embodiments, the expression vector further comprises a gene that stabilizes large plasmids. In

some embodiments, the expression vector further comprises a sopA gene, a sopB gene, and/or a sopC gene.

In some embodiments, the expression vector is configured to accept an insert comprising more than 10 kb, more than 20 kb, more than 50 kb, or more than 100 kb (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the expression vector is configured to accept an insert comprising 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the expression vector is configured to accept a biosynthetic gene cluster nucleic acid at the cloning site and express a product of the biosynthetic gene cluster nucleic acid under control of the first promoter and/or the second promoter. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid when said vector is present in a host cell. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid when said vector is present in a *Streptomyces* host cell. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid that is a nucleic acid or a polypeptide. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid that is a product of one or more enzymes encoded by the biosynthetic gene cluster. In some embodiments, the expression vector is configured to express a product that is a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomatic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a product of a polyketide synthase or nonribosomal peptide synthase.

In some embodiments, the technology provides a kit. For example, in some embodiments, the technology provides a kit comprising an expression vector as described herein. In some embodiments, the first promoter of the expression vector is an inducible promoter and/or the second promoter of the expression vector is an inducible promoter and the kit further comprises an inducer of said first promoter and/or an inducer of said second promoter. In some embodiments, the kit further comprises a restriction enzyme for cutting said expression vector at said cloning site and/or a composition for integrating a nucleic acid at said cloning site.

In some embodiments, the technology provides a system. For example, in some embodiments, the technology provides a system comprising an expression vector as described herein. In some embodiments, the first promoter of the expression vector is an inducible promoter and/or the second promoter of the expression vector is an inducible promoter and the system further comprises an inducer of said first promoter and/or an inducer of said second promoter. In some embodiments, the system further comprises a restriction enzyme for cutting said expression vector at set cloning site and/or a composition for integrating a nucleic acid at said cloning site. In some embodiments, the system further

comprises a culture medium, an antibiotic, and/or a competent host for said expression vector.

In some embodiments, the technology provides an expression vector as described herein that further comprises an insert (e.g., a cloned insert). Accordingly, in some embodiments, the technology provides a nucleic acid comprising an expression vector and an insert, wherein said expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and the insert comprises a biosynthetic gene cluster nucleic acid. In some embodiments, the insert is 5 kb or more, 10 kb or more, and/or 20, 30, 40, 50, 60, 70, 80, 90, or 100 kb or more. In some embodiments, the insert is at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the expression vector is configured to accept an insert comprising 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)).

In some embodiments, the insert is from a cultured microorganism. In some embodiments, the insert is from a metagenomic library. In some embodiments, the insert comprises a nucleotide sequence encoding an amino acid sequence of a polyketide synthase (PKS) or a nonribosomal peptide synthase (NRPS). In some embodiments, the insert comprises a plurality of genes. In some embodiments, the insert comprises genes encoded by both strands of said insert. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid under control of the first promoter and/or the second promoter. In some embodiments, the first promoter and/or the second promoter is an inducible promoter. In some embodiments, the first promoter and/or the second promoter directs transcription in a host cell that is different than the source of the insert. In some embodiments, the first promoter and/or the second promoter directs transcription in *Streptomyces*. In some embodiments, the first promoter is Potr or Potr*. In some embodiments, the second promoter is PnitA. In some embodiments, first promoter comprises a nucleotide sequence provided by SEQ ID NO: 1. In some embodiments, the first promoter comprises a nucleotide sequence provided by SEQ ID NO: 2. In some embodiments, the second promoter comprises a nucleotide sequence provided by SEQ ID NO: 6. In some embodiments, the expression vector further comprises OtrR. In some embodiments, the expression vector further comprises NitR. In some embodiments, the expression vector further comprises a nucleotide sequence provided by SEQ ID NO: 3. In some embodiments, the expression vector further comprises a nucleotide sequence provided by SEQ ID NO: 5.

In some embodiments, the vector further comprises a selectable marker for *Streptomyces*. In some embodiments, the expression vector further comprises a selectable marker for *E. coli*. In some embodiments, the expression vector further comprises an *E. coli* origin of replication. In some embodiments, the expression vector further comprises a *Streptomyces* origin of replication. In some embodiments, the expression vector further comprises a gene that stabilizes large plasmids. In some embodiments, expression vector further comprises a sopA gene, a sopB gene, and/or a sopC gene. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster

nucleic acid when said vector is present in a host cell. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid when said vector is present in a *Streptomyces* host cell.

In some embodiments, the biosynthetic gene cluster nucleic acid comprises a nucleotide sequence encoding two or more genes of a biosynthetic pathway. In some embodiments, the biosynthetic gene cluster nucleic acid comprises a nucleotide sequence encoding two or more genes of a biosynthetic pathway that produces a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a product of a polyketide synthase or nonribosomal peptide synthase.

In some embodiments, the technology provides a nucleic acid comprising an expression vector and an insert, wherein the expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and the insert comprises at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the expression vector is configured to accept an insert comprising 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the first promoter and/or the second promoter is an inducible promoter. In some embodiments, the first promoter and/or the second promoter directs transcription in *Streptomyces*. In some embodiments, the first promoter is Potr or Potr*. In some embodiments, the second promoter is PnitA. In some embodiments, the first promoter comprises a nucleotide sequence provided by SEQ ID NO: 1. In some embodiments, the first promoter comprises a nucleotide sequence provided by SEQ ID NO: 2. In some embodiments, the second promoter comprises a nucleotide sequence provided by SEQ ID NO: 6. In some embodiments, the expression vector further comprises OtrR. In some embodiments, the expression vector further comprises NitR. In some embodiments, the expression vector further comprises a nucleotide sequence provided by SEQ ID NO: 3. In some embodiments, the expression vector further comprises a nucleotide sequence provided by SEQ ID NO: 5. In some embodiments, the expression vector further comprises a selectable marker for *Streptomyces*. In some embodiments, the expression vector further comprises a selectable marker for *E. coli*. In some embodiments, the expression vector further comprises an *E. coli* origin of replication. In some embodiments, the expression vector further comprises a *Streptomyces* origin of replication. In some embodiments, the expression vector further comprises a gene that stabilizes large plasmids. In some embodiments, the expression vector further comprises a sopA gene, a sopB gene, and/or a sopC gene. In some embodiments, the insert comprises more than 10 kb, more than 20 kb, more than 50 kb, or more than 100 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110,

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115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the expression vector is configured to accept an insert comprising 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)).

In some embodiments, the insert comprises a biosynthetic gene cluster. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster when said nucleic acid is present in a host cell. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid when said vector is present in a *Streptomyces* host cell. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid that is a nucleic acid or a polypeptide. In some embodiments, the expression vector is configured to express a product of the biosynthetic gene cluster nucleic acid that is a product of one or more enzymes encoded by the biosynthetic gene cluster. In some embodiments, the biosynthetic gene cluster nucleic acid comprises a nucleotide sequence encoding two or more genes of a biosynthetic pathway that produces a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a product of a polyketide synthase or nonribosomal peptide synthase.

In some embodiments, the technology provides a host cell comprising an expression vector as described herein. In some embodiments, the technology provides a host comprising an expression vector further comprising an insert. In some embodiments, the technology provides a host cell comprising a nucleic acid, wherein said nucleic acid comprises an expression vector as described herein and a nucleic acid insert. In some embodiments, the host cell expresses a product of a biosynthetic gene cluster encoded by the insert. In some embodiments, the host cell expresses a nucleic acid or a polypeptide encoded by the insert. In some embodiments, the host cell expresses one or more enzymes encoded by the insert. In some embodiments, the host cell expresses a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a polyketide or nonribosomal peptide.

In some embodiments, the technology provides a composition comprising a host cell as described herein. For example, in some embodiments, the technology provides a composition comprising a host cell as described herein, wherein the host cell comprises an expression vector as described herein (e.g., comprising an insert) and the com-

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position further comprises an inducer of the first promoter and/or an inducer of the second promoter. In some embodiments, the composition further comprises a product expressed from the induced expression of said insert. In some embodiments, the product is a biologically active agent. In some embodiments, the biologically active agent is a polyketide or nonribosomal peptide. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a terpene, saccharide, or alkaloid.

In some embodiments, the technology provides methods. For example, in some embodiments, the technology provides a method of expressing a product from a cloned biosynthetic gene cluster. In some embodiments, methods comprise providing an expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and cloning a nucleic acid insert comprising a biosynthetic gene cluster at said cloning site. In some embodiments, the nucleic acid insert is 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, or 100 kb or more (e.g., comprising more than 10 kb, more than 20 kb, more than 50 kb, and/or more than 100 kb (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)).

In some embodiments, the technology provides methods of expressing a product from a cloned nucleic acid insert comprising at least 10 kb (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, methods comprise providing an expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and cloning a nucleic acid insert comprising at least 10 kb at said cloning site. In some embodiments, the nucleic acid insert comprises a biosynthetic gene cluster.

In some embodiments, the technology provides methods for expressing a product from a cloned biosynthetic gene cluster. In some embodiments, methods comprise providing an expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; cloning a nucleic acid insert comprising a biosynthetic gene cluster at said cloning site to provide a recombinant vector comprising said nucleic acid insert; transforming said recombinant vector compris-

ing said nucleic acid insert into a host cell; and contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said nucleic acid insert. In some embodiments, the host cell is a *Streptomyces* spp. In some embodiments, the nucleic acid insert is 5 kb or more, 10 kb or more, or 20, 30, 40, 50, 60, 70, 80, 90, or 100 kb or more (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the nucleic acid insert is from a cultured microorganism. In some embodiments, the nucleic acid insert is from a metagenomic library. In some embodiments, the nucleic acid insert comprises a nucleotide sequence encoding a polyketide synthase (PKS) or a nonribosomal peptide synthase (NRPS). In some embodiments, the nucleic acid insert comprises a plurality of genes. In some embodiments, the nucleic acid insert comprises genes encoded by both strands of said nucleic acid insert. In some embodiments, the methods further comprise detecting expression of a product encoded by one or more nucleotide sequences of said nucleic acid insert. In some embodiments, the product is produced by a biosynthetic pathway encoded by the biosynthetic gene cluster. In some embodiments, the product is a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a polyketide or nonribosomal peptide. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a terpene, saccharide, or alkaloid.

In some embodiments, the technology provides methods for expressing a product from a biosynthetic gene cluster. For example, in some embodiments, methods comprise providing a host cell comprising a recombinant nucleic acid comprising an expression vector and an insert, wherein said expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and said insert comprises a biosynthetic gene cluster nucleic acid; and contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said biosynthetic gene cluster. In some embodiments, the host cell is a *Streptomyces* spp. In some embodiments, the insert is 5 kb or more, 10 kb or more, or 20, 30, 40, 50, 60, 70, 80, 90, or 100 kb or more (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the insert is from a cultured

microorganism. In some embodiments, the insert is from a metagenomic library. In some embodiments, the insert comprises a nucleotide sequence encoding a polyketide synthase (PKS) or a nonribosomal peptide synthase (NRPS). In some embodiments, the insert comprises a plurality of genes. In some embodiments, the insert comprises genes encoded by both strands of said insert. In some embodiments, the methods further comprise detecting expression of a product encoded by the biosynthetic gene cluster. In some embodiments, the product is a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a polyketide or nonribosomal peptide. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a terpene, saccharide, or alkaloid.

In some embodiments, the technology provides methods for expressing a product from a nucleic acid insert comprising at least 10 kb (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, methods comprise providing a host cell comprising a recombinant nucleic acid comprising an expression vector and an insert, wherein said expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and said insert comprises at least 10 kb (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)); and contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said nucleic acid insert. In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the host cell is a *Streptomyces* spp. In some embodiments, the insert comprises a biosynthetic gene cluster. In some embodiments, the insert is from a cultured microorganism. In some embodiments, the insert is from a metagenomic library. In some embodiments, the insert comprises a nucleotide sequence encoding a polyketide synthase (PKS) or a nonribosomal peptide synthase (NRPS). In some embodiments, the insert comprises a plurality of genes. In some embodiments, the insert comprises genes encoded by both strands of said insert. In some embodiments, methods further comprise detecting expression of said product. In some embodiments, the product is a biologically active agent. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a polyketide or nonribosomal peptide. In some embodiments, the biologi-

cally active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent. In some embodiments, the biologically active agent is a terpene, saccharide, or alkaloid.

In some embodiments, the technology provides methods for identifying a nucleic acid comprising a biosynthetic gene cluster. For example, in some embodiments, methods comprise providing a host cell comprising a recombinant nucleic acid comprising an expression vector and an insert, wherein said expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and the expression vector is configured to express a product of the insert under control of the first promoter and/or the second promoter; contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said insert; detecting expression of said product; and identifying the nucleic acid as a nucleic acid comprising a biosynthetic gene cluster when said product is identified. In some embodiments, the host cell is a *Streptomyces* spp. In some embodiments, the insert is 5 kb or more, 10 kb or more, and/or 20, 30, 40, 50, 60, 70, 80, 90, or 100 kb or more (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the insert is from a cultured microorganism. In some embodiments, the insert is from a metagenomic library. In some embodiments, the product is a biologically active agent. In some embodiments, the biologically active agent is a polyketide or a nonribosomal peptide. In some embodiments, the biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative agent.

In some embodiments, the detecting step of the method comprises a selection or a screen.

In some embodiments, the technology provides a library of nucleic acids, e.g., a library of expression vectors as described herein comprising nucleic acid inserts, wherein the library comprises a plurality of different insert nucleotide sequences (e.g., a library of cloned inserts in the expression vector). In some embodiments, the technology provides a clone library in host cells. In some embodiments, the technology provides a library comprising a plurality of host cells as described herein (e.g., host cells comprising a library of expression vectors as described herein comprising nucleic acid inserts, wherein the library comprises a plurality of different insert nucleotide sequences (e.g., a library of hosts comprising cloned inserts in the expression vector).

In some embodiments, the technology finds use to express a product from a biosynthetic gene cluster, the expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein said expression vector is configured to accept a biosynthetic gene cluster nucleic acid at the cloning site and express a product of the biosynthetic gene cluster nucleic acid under control of the first promoter and/or the second promoter. In some embodiments, the technology finds use to express a product from a cloned nucleic acid insert comprising at least 10 kb (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)), the expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein said expression vector is configured to accept a nucleic acid comprising at least 10 kb (e.g., at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)) at the cloning site and express a product of the nucleic acid under control of the first promoter and/or the second promoter. In some embodiments, the nucleic acid insert is 10-200 kb (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)). In some embodiments, the technology finds use to identify a nucleic acid comprising a biosynthetic gene cluster, the expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein said expression vector is configured to accept a nucleic acid at the cloning site and express a product of the nucleic acid under control of the first promoter and/or the second promoter. In some embodiments, the technology finds use to produce a bioactive agent produced by a biosynthetic gene cluster, the expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein said expression vector is configured to accept a nucleic acid comprising a biosynthetic gene cluster nucleic acid at the cloning site and express a product of the biosynthetic gene cluster under control of the first promoter and/or the second promoter.

The technology also relates to nucleic acids. For example, in some embodiments, the technology provides a nucleic acid comprising a nucleotide sequence provided by SEQ ID NO: 8. In some embodiments, the technology provides a nucleic acid comprising a nucleotide sequence provided by SEQ ID NO: 9.

Additional embodiments will be apparent to persons skilled in the relevant art based on the teachings contained herein.

Definitions

To facilitate an understanding of the present technology, a number of terms and phrases are defined below. Additional definitions are set forth throughout the detailed description.

Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless

the context clearly dictates otherwise. The phrase “in one embodiment” as used herein does not necessarily refer to the same embodiment, though it may. Furthermore, the phrase “in another embodiment” as used herein does not necessarily refer to a different embodiment, although it may. Thus, as described below, various embodiments of the technology may be readily combined, without departing from the scope or spirit of the technology.

In addition, as used herein, the term “or” is an inclusive “or” operator and is equivalent to the term “and/or” unless the context clearly dictates otherwise. The term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a”, “an”, and “the” include plural references. The meaning of “in” includes “in” and “on.”

As used herein, the terms “about”, “approximately”, “substantially”, and “significantly” are understood by persons of ordinary skill in the art and will vary to some extent on the context in which they are used. If there are uses of these terms that are not clear to persons of ordinary skill in the art given the context in which they are used, “about” and “approximately” mean plus or minus less than or equal to 10% of the particular term and “substantially” and “significantly” mean plus or minus greater than 10% of the particular term.

As used herein, disclosure of ranges includes disclosure of all values and further divided ranges within the entire range, including endpoints and sub-ranges given for the ranges.

As used herein, the suffix “-free” refers to an embodiment of the technology that omits the feature of the base root of the word to which “-free” is appended. That is, the term “X-free” as used herein means “without X”, where X is a feature of the technology omitted in the “X-free” technology. For example, a “calcium-free” composition does not comprise calcium, a “mixing-free” method does not comprise a mixing step, etc.

Although the terms “first”, “second”, “third”, etc. may be used herein to describe various steps, elements, compositions, components, regions, layers, and/or sections, these steps, elements, compositions, components, regions, layers, and/or sections should not be limited by these terms, unless otherwise indicated. These terms are used to distinguish one step, element, composition, component, region, layer, and/or section from another step, element, composition, component, region, layer, and/or section. Terms such as “first”, “second”, and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first step, element, composition, component, region, layer, or section discussed herein could be termed a second step, element, composition, component, region, layer, or section without departing from technology.

As used herein, an “increase” or a “decrease” refers to a detectable (e.g., measured) positive or negative change in the value of a variable relative to a previously measured value of the variable, relative to a pre-established value, and/or relative to a value of a standard control. An increase is a positive change preferably at least 10%, more preferably 50%, still more preferably 2-fold, even more preferably at least 5-fold, and most preferably at least 10-fold relative to the previously measured value of the variable, the pre-established value, and/or the value of a standard control. Similarly, a decrease is a negative change preferably at least 10%, more preferably 50%, still more preferably at least 80%, and most preferably at least 90% of the previously measured value of the variable, the pre-established value,

and/or the value of a standard control. Other terms indicating quantitative changes or differences, such as “more” or “less,” are used herein in the same fashion as described above.

Unless otherwise defined herein, scientific and technical terms used in connection with the present technology shall have the meanings that are commonly understood by those of ordinary skill in the art. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. Generally, nomenclatures used in connection with, and techniques of, cell and tissue culture, molecular biology, immunology, microbiology, genetics, and protein and nucleic acid chemistry and hybridization described herein are those well-known and commonly used in the art. The methods and techniques of the present technology are generally performed according to conventional methods well known in the art and as described in various general and more specific references that are cited and discussed throughout the present specification unless otherwise indicated. See, e.g., Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 2d ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (1989); Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 3d ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2000); Ausubel et al., *Current Protocols in Molecular Biology*, Greene Publishing Associates (1992 and Supplements to 2000); Ausubel et al., *Short Protocols in Molecular Biology: A Compendium of Methods from Current Protocols in Molecular Biology*, 4th ed., Wiley & Sons (1999); Harlow and Lane, *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (1990); Harlow and Lane, *Using Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (1998); and T. Kieser et al., *Practical Streptomyces Genetics*, John Innes Foundation, Norwich (2000); each of which is incorporated herein by reference in its entirety.

Unless specifically defined or described in a different way elsewhere herein, the following terms and descriptions related to the technology shall be understood as given below.

As used herein, the term “recombinant”, when used with reference, e.g., to a cell, nucleic acid, protein, or vector, indicates that the cell, nucleic acid, protein, or vector has been modified by the introduction of a heterologous nucleic acid or protein or the alteration of a native nucleic acid or protein, or that the material is derived from a cell so modified. Thus, e.g., recombinant cells express genes that are not found within the native (non-recombinant) form of the cell or express native genes that are otherwise abnormally expressed, under expressed, or not expressed at all.

As used herein, the term “vector” refers to a DNA molecule used as a vehicle to carry a nucleic acid (e.g., an “insert” comprising foreign genetic material) into a cell where, in some embodiments, it is replicated and/or expressed.

As used herein, the term “configured to accept” or “adapted to accept” refers to the characteristic (e.g., by design, structure, and/or function) of a nucleic acid (e.g., a vector (e.g., an expression vector)) as being capable of being linked (e.g., by ligation, recombination, or other nucleic acid manipulation in vivo and/or in vitro) to another nucleic acid. For example, a vector “configured to accept” an insert means that the vector is capable of being joined with the insert to provide a recombinant nucleic acid comprising the vector and the insert. In some embodiments, a vector that is “configured to accept” an insert comprises a cloning site and the cloning site comprises a restriction enzyme recognition

site, a site of homologous recombination, or other nucleic acid structure or sequence that facilitates introduction of an insert at the cloning site. In some embodiments, a vector is prepared to receive an insert nucleic acid by digesting the vector and a nucleic acid that comprises the insert with restriction enzymes. The digested nucleic acids are then spliced together by an enzyme called ligase (e.g., by a process known as ligation) to form a recombinant vector capable of expressing a nucleic acid of the insert. In some embodiments, "TA" cloning is used to produce a recombinant vector from a vector and an insert. In some embodiments, a vector comprises sites for in vitro recombination reactions (e.g., integration and excision) similar to those that occur when lambda phage infects bacteria. In some embodiments, the recombination reactions are facilitated by the recombination of attP and attB attachment sites directed by a "clonase" or "integrase" enzyme. Other strategies are known in the art for designing and producing a cloning site that is configured to accept an insert.

As used herein, the term "expression vector" refers to a vector used to introduce a specific nucleic acid into a target cell for expression of the nucleic acid by the cell, e.g., to produce one or more proteins encoded by the nucleic acid by a constitutive or an inducible promoter. In some embodiments, an expression vector is a nucleic acid comprising one or more promoters. In some embodiments, an expression vector comprises one or more convenient restriction sites to allow for insertion or substitution of a nucleic acid into the expression vector.

As used herein, "expression" refers to the process by which the information of a particular nucleic acid (e.g., a gene) is used to synthesize a product (e.g., a biomolecule (e.g., a nucleic acid, a polypeptide, a carbohydrate, a lipid, and combinations, derivatives, and/or metabolites of the foregoing); a metabolite (e.g., a primary metabolite, a secondary metabolite); a fatty acid; a polyketide; a nucleotide; an amino acid; a cofactor; and combinations, derivatives, and/or metabolites of the foregoing). The term "expression" includes but is not limited to one or more of the following: transcription of a gene into a precursor mRNA; processing of a precursor mRNA to produce a mature mRNA; mRNA stability; translation of a mature mRNA into a protein (including codon usage and tRNA availability); and/or glycosylation and/or other modifications of the translation product. The term "expression" also includes transcription of a non-coding RNA, e.g., a transfer RNA, a ribosomal RNA, a microRNAs, a siRNA, a piRNA, a snoRNA, a snRNA, an exRNA, a scaRNA, or a long ncRNA. The term "expression" includes production of a functional product and production of non-functional products that find use in producing functional products by subsequent chemical or biochemical manipulation or synthesis.

As used herein, the term "biologically active agent" refers to any substance that has activity in a biological system and/or organism. For instance, in some embodiments, a "biologically active agent" is a substance that, when administered to an organism, has a biological effect on that organism. In some embodiments, where a substance is biologically active, a portion of that substance that shares at least one biological activity of the whole substance is typically referred to as a "biologically active" portion. In some embodiments, a "biologically active agent" is a chemical substance or formulation that beneficially affects humans, animals, or plants or is intended for use in the cure, mitigation, treatment, prevention, or diagnosis of infection or disease, or is destructive to or inhibits the growth of microorganisms.

As used herein the term "exogenously" refers to the use of native or non-native promoters that are outside the boundaries of a cloned nucleic acid insert, e.g., flanking the outside boundaries of a cloned BGC.

As used herein the term "endogenously" refers to the use of native promoters within the boundaries of a cloned nucleic acid insert, e.g., within the boundaries of a cloned BGC.

As used herein, the term "shuttle vector" refers to a vector constructed so that it can propagate in two different host species, e.g., *E. coli* and another organism such as *Streptomyces*.

As used herein, the term "promoter" refers to a region of a nucleic acid that controls the binding of RNA polymerase and transcription factors (e.g., the sequence of the promoter region controls the binding of the RNA polymerase and/or transcription factors). In some embodiments, a promoter drives transcription of a target gene or genes and thus may determine the timing and/or amount of gene expression and determines the amount of a recombinant protein that is produced. The term "promoter" may refer to a combination of a promoter (e.g., the RNA polymerase binding site) and an operator (e.g., response elements). Promoters typically comprise approximately 100 to 1000 base pairs and are present upstream of their target genes. Many common promoters are always active and are thus referred to as constitutive promoters. Other promoters are only active under specific circumstances and are thus referred to as "inducible promoters", which can be switched between two discrete states, e.g., an OFF state and an ON state. Some inducible promoters provide control of expression over a continuous range that is a function of the amount of inducer provided and/or present.

As used herein, "inducible promoter" means that the recognition of the promoter by the RNA polymerase, and therefore the transcriptional activity of the promoter and its target gene, is controlled by the absence, presence, or amount of chemical or physical factors.

As used herein, the term "DNA transcription" refers to the process of synthesizing a RNA from a DNA molecule by a specialized enzyme that is an RNA polymerase.

As used herein, the term "constitutive gene" or "constitutively expressed gene" refers to a gene that is transcribed continually at a relatively constant level. This term implies that a constitutive promoter regulates DNA transcription for the gene and therefore that an encoded gene product (e.g., protein or RNA) is produced at a relatively constant level.

As used herein, "ribosome binding site" refers to an RNA sequence to which ribosomes can bind to initiate protein synthesis (translation) inside a host cell or organism as part of the process of expressing a protein, a product produced by the protein, and/or a product produced by a biosynthetic pathway in which the protein is a member.

As used herein, "foreign gene expression" means the entire process by which the information of a particular gene or biosynthetic pathway is used to synthesize a product in a heterologous host. As used herein in reference to gene expression, the term "foreign" means that the referenced gene is from an organism different than the host used for gene expression.

As used herein, "outward-reading" refers to the direction of transcription from a specific promoter that is located particularly within a defined region of a DNA and that is typically located near the 5' or 3' ends of the defined region of the DNA. In particular, "outward-reading" refers to transcription from the mentioned promoter at which RNA synthesis starts within the defined region of the DNA and

that proceeds towards a boundary of the defined region of the DNA toward adjacent DNA.

As used herein, a biosynthetic gene cluster (BGC) can be defined as a physically clustered group of two or more genes in a particular genome that together encode a biosynthetic pathway for the production of a specialized metabolite and/or chemical variants thereof. Non-limiting exemplary BGCs encode multiple genes for biosynthetic pathways that produce polyketides, nonribosomal peptides (NRPs), ribosomally synthesized and post-translationally modified peptides (RiPPs), terpenes, saccharides, and alkaloids. Some BGCs comprise elements such as acyltransferase domain substrate specificities and starter units for polyketide BGCs, release/cyclization types and adenylation domain substrate specificities for NRP BGCs, precursor peptides and peptide modifications for RiPP BGCs, glycosyltransferase specificities for saccharide BGCs, and hybrids combining one or more of these units.

As used herein, the term “natural product” refers to biological products that can be found in nature. Embodiments of the technology disclosed herein find use as effective tools to discover unknown natural products.

As used herein, the term “small molecule” or “metabolite” refers to a composition that has a molecular weight of less than approximately 5 kDa and more preferably less than approximately 2 kDa. Small molecules can be, e.g., nucleic acids, peptides, polypeptides, glycopeptides, peptidomimetics, carbohydrates, lipids, antibiotics, lipopolysaccharides, fatty acids, polyketides, nucleotides, amino acids, cofactors, and combinations, derivatives, and/or metabolites of the foregoing, or other organic or inorganic molecules.

As used herein the terms “polyketide” and “nonribosomal peptide” refer to important classes of natural products. Polyketides are a large group of secondary metabolites that either comprise alternating carbonyl and methylene groups or are derived from precursors that comprise such alternating groups. Non-ribosomal peptides have a chemical structure similar to proteins (e.g., comprise peptide bonds), but are biosynthesized without use of messenger RNA.

As used herein, the term “polyketide synthase” or “PKS” refers to a protein with modular enzymatic activities that can lead to production of a polyketide under certain conditions.

As used herein, the term “non-ribosomal peptide synthetase” or “NRPS” refers to a protein with modular enzymatic activities that can lead to production of a non-ribosomal peptide under certain conditions.

As used herein, the term “ribosomally synthesized and/or post-translationally modified polypeptide (RiPP)” refers to genetically encoded precursor peptides that undergo some degree of enzymatic post-translational modification (e.g., chemical transformations occurring after translation).

As used herein, the term “module” refers to a section of a polyketide synthase or non-ribosomal peptide synthetase protein comprising one or more domains and involved in at least one round (typically one round) of chain extension or chain transfer (more commonly chain extension), including but not limited to a ketosynthase, ketoreductase, dehydratase, enoyl reductase, acyl carrier protein, acyl transferase, thioesterase, condensation, thiolation, peptidyl carrier protein, methylation or adenylation domain.

As used herein, the term “microorganism” includes prokaryotic and eukaryotic microbial species from the domains Archaea, Bacteria, and Eukarya, the latter including yeast and filamentous fungi, protozoa, algae, or higher Protista. The terms “microbial cells” and “microbes” are used interchangeably with the term “microorganism”.

The terms “bacteria” and “bacterium” and “archaea” and “archaeon” refer to prokaryotic organisms of the domain Bacteria and Archaea in the three-domain system (see Woese C. R., et al., Proc Natl Acad Sci USA 1990, 87:4576-79).

The term “Archaea” refers to a taxonomic domain of organisms typically found in unusual environments and distinguished from the rest of the prokaryotes by several criteria, including the number of ribosomal proteins and the lack of muramic acid in cell walls. On the basis of small subunit rRNA analysis, the Archaea consist of two phylogenetically-distinct groups: Crenarchaeota and Euryarchaeota. On the basis of their physiology, the Archaea can be organized into three types: methanogens (prokaryotes that produce methane); extreme halophiles (prokaryotes that live at very high concentrations of salt (NaCl)); and extreme (hyper) thermophiles (prokaryotes that live at very high temperatures). Besides the unifying archaeal features that distinguish them from Bacteria (e.g., no murein in cell wall, ester-linked membrane lipids, etc.), these prokaryotes exhibit unique structural or biochemical attributes which adapt them to their particular habitats. The Crenarchaeota consist mainly of hyperthermophilic sulfur-dependent prokaryotes and the Euryarchaeota contain the methanogens and extreme halophiles.

The term “Bacteria” or “eubacteria” refers to a domain of prokaryotic organisms. Bacteria include at least 11 distinct groups as follows: (1) Gram-positive (gram+) bacteria, of which there are two major subdivisions: (1) high G+C group (Actinomycetes, Mycobacteria, *Micrococcus*, others) (2) low G+C group (*Bacillus*, *Clostridia*, *Lactobacillus*, Staphylococci, Streptococci, Mycoplasmas); (2) Proteobacteria, e.g., Purple photosynthetic+non-photosynthetic Gram-negative bacteria (includes most “common” Gram-negative bacteria); (3) Cyanobacteria, e.g., oxygenic phototrophs; (4) Spirochetes and related species; (5) *Planctomyces*; (6) *Bacteroides*, Flavobacteria; (7) *Chlamydia*; (8) Green sulfur bacteria; (9) Green non-sulfur bacteria (also anaerobic phototrophs); (10) Radioresistant micrococci and relatives; (11) *Thermotoga* and *Thermosipho thermophiles*.

“Gram-negative bacteria” include cocci, nonenteric rods, and enteric rods. The genera of Gram-negative bacteria include, for example, *Neisseria*, *Spirillum*, *Pasteurella*, *Brucella*, *Yersinia*, *Francisella*, *Haemophilus*, *Bordetella*, *Escherichia*, *Salmonella*, *Shigella*, *Klebsiella*, *Proteus*, *Vibrio*, *Pseudomonas*, *Bacteroides*, *Acetobacter*, *Aerobacter*, *Agrobacterium*, *Azotobacter*, *Spirilla*, *Serratia*, *Rhizobium*, *Chlamydia*, *Rickettsia*, *Treponema*, and *Fusobacterium*.

“Gram positive bacteria” include cocci, nonsporulating rods, and sporulating rods. The genera of gram positive bacteria include, for example, *Actinomyces*, *Bacillus*, *Clostridium*, *Corynebacterium*, *Erysipelothrix*, *Lactobacillus*, *Listeria*, *Mycobacterium*, *Myxococcus*, *Nocardia*, *Staphylococcus*, *Streptococcus*, and *Streptomyces*.

The term “genus” is defined as a taxonomic group of related species according to the Taxonomic Outline of Bacteria and Archaea (Garrrity et al. (2007) The Taxonomic Outline of Bacteria and Archaea. TOBA Release 7.7, March 2007. Michigan State University Board of Trustees).

The term “species” is defined as a collection of closely related organisms with greater than 97% 16S ribosomal RNA sequence homology and greater than 70% genomic hybridization and sufficiently different from all other organisms so as to be recognized as a distinct unit.

The term “strain” as used herein in reference to a microorganism describes an isolate of a microorganism considered to be of the same species but with a unique genome and, if nucleotide changes are non-synonymous, a unique pro-

teome differing from other strains of the same organism. Strains may differ in their non-chromosomal genetic complement. Typically, strains are the result of isolation from a different host or at a different location and time, but multiple strains of the same organism may be isolated from the same host.

As used herein, the term “naturally occurring” as applied to a nucleic acid, an enzyme, a cell, or an organism, refers to a nucleic acid, enzyme, cell, or organism that is found in nature. For example, a polypeptide or polynucleotide sequence that is present in an organism that can be isolated from a source in nature and that has not been intentionally modified by a human in the laboratory is naturally occurring.

As used herein, the term “non-naturally occurring” as applied to a nucleic acid, an enzyme, a cell, or an organism refers to a nucleic acid, an enzyme, a cell, or an organism that has at least one genetic alteration not normally found in the naturally occurring nucleic acid, enzyme, cell, or organism. Genetic alterations include, for example, modifications introducing expressible nucleic acids encoding metabolic polypeptides, other nucleic acid additions, nucleic acid deletions, and/or other functional disruption of the microbial genetic material. Such modifications include, for example, coding regions and functional fragments thereof, for heterologous, homologous, or both heterologous and homologous polypeptides for the referenced species. Additional modifications include, for example, non-coding regulatory regions in which the modifications alter expression of a gene or operon.

As used herein, the term “host cell”, “host microbial organism”, and “host microorganism” are used interchangeably to refer to any archaeal, bacterial, or eukaryotic living cell into which a heterologous entity (e.g., a biomolecule such as a nucleic acid, protein, etc.) can be, or has been, inserted. The term also relates to the progeny of the original cell, which may not be completely identical in morphology or in genomic or total DNA complement to the original parent, due to natural, accidental, or deliberate mutation.

The terms “modified microorganism,” “recombinant microorganism”, and “recombinant host cell” are refer to a non-naturally occurring organism that is produced by methods such as inserting, expressing, or overexpressing endogenous polynucleotides; by expressing or overexpressing heterologous polynucleotides, such as those included in an integrated and/or episomal vector; by introducing a mutation into the microorganism; or by altering the expression of an endogenous gene. In embodiments relating to the introduction of a polynucleotide into a microorganism, the polynucleotide generally encodes a one or more enzymes involved in a biosynthetic pathway for producing a desired metabolite. It is understood that the terms “recombinant microorganism” and “recombinant host cell” refer not only to the particular recombinant microorganism but to the progeny or potential progeny of such a microorganism. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term as used herein.

The term “wild-type microorganism” describes a cell that occurs in nature, e.g., a cell that has not been genetically modified. A wild-type microorganism can be genetically modified to express or overexpress a target enzyme. This microorganism can act as a parental microorganism in the generation of a microorganism modified to express or overexpress a target enzyme. In turn, the microorganism modi-

fied to express or overexpress one or more target enzymes can be modified to express or overexpress another target enzyme.

Accordingly, a “parental microorganism” functions as a reference cell for successive genetic modification events. Each modification event can be accomplished by introducing a nucleic acid molecule into the reference cell. The introduction facilitates the expression or overexpression of a target enzyme. It is understood that the term “facilitates” encompasses the activation of endogenous polynucleotides encoding a target enzyme through genetic modification of, e.g., a promoter sequence in a parental microorganism. It is further understood that the term “facilitates” encompasses the introduction of heterologous polynucleotides encoding a target enzyme in to a parental microorganism.

The term “mutation” as used herein indicates any modification of a nucleic acid that results in an altered nucleic acid, e.g., that produces an amino acid “substitution” in a polypeptide (e.g., thus producing a “mutant” polypeptide or “mutant” nucleic acid). Mutations include, for example, point mutations, deletions, or insertions of single or multiple residues in a polynucleotide, which includes alterations arising within a protein-encoding region of a gene as well as alterations in regions outside of a protein-encoding sequence, such as, but not limited to, regulatory or promoter sequences. A genetic alteration may be a mutation of any type. For instance, the mutation may constitute a point mutation, a frame-shift mutation, an insertion, or a deletion of part or all of a gene. In addition, in some embodiments of the modified microorganism, a portion of the microorganism genome has been replaced with a heterologous polynucleotide. In some embodiments, the mutations are naturally-occurring. In other embodiments, the mutations are the results of artificial mutation pressure. In still other embodiments, the mutations in the microorganism genome are the result of genetic engineering.

The term “biosynthetic pathway”, also referred to as “metabolic pathway”, refers to a set of anabolic or catabolic biochemical reactions for converting one chemical species into another. Gene products belong to the same “biosynthetic pathway” if they, in parallel or in series, act on the same substrate, produce the same product, or act on or produce a metabolic intermediate (e.g., a metabolite) between the same substrate and metabolite end product.

The term “gene module” refers to a group, set, or collection of genes, e.g., as provided in a BGC. In some embodiments, an operon comprises the genes of a gene module; in some embodiments, the genes of a gene module belong to the same biosynthetic pathway. In some embodiments, a BGC comprises the genes of a gene module. In some embodiments, the genes of a gene module are coexpressed, e.g., the same set of transcription factors binds to the genes of the gene module to modulate expression of the genes of the gene module. In some embodiments, the genes of a gene module are provided together on a nucleic acid. In some embodiments, the genes are provided together on a nucleic acid in the same arrangement as found in nature and, in some embodiments, the genes of the gene module are provided on a nucleic acid in an arrangement that is not found in nature. In some embodiments, a gene module comprises a novel group, set, or collection of genes that are not normally present in the same pathway in nature. In some embodiments, a gene module comprises a novel group, set, or collection of genes that are not normally present in the same organism in nature.

The term “heterologous” as used herein with reference to molecules and in particular enzymes and polynucleotides,

indicates molecules that are expressed in an organism other than the organism from which they originated or are found in nature, independently of the level of expression, which can be lower, equal, or higher than the level of expression of the molecule in the native microorganism.

As used herein, the term "heterologous host" refers to a host organism, usually a bacterial strain, that can express one or more genes from another organism (e.g., "source organism") that is taxonomically classified as belonging to a different genus or species than the host organism. The "heterologous host" has the potential to express a product of the one or more genes from the other organism (e.g., "source organism") when cultured under appropriate conditions.

As used herein, the term "homologous host" refers to an organism, usually a bacterial strain, that can express one or more genes from an identical or essentially identical organism. The "homologous host" has the potential to express a product of the one or more genes when cultured under appropriate conditions.

As used herein the term "*Streptomyces* expression strains" or "heterologous *Streptomyces* expression strains" refers to bacterial strains including, but not limited to, commonly used species such as *Streptomyces avermitilis*, *Streptomyces venezuelae*, *Streptomyces albus*, *Streptomyces lividans*, and *Streptomyces coelicolor*.

As used herein, the term "*Streptomyces* spp." refers to any strain including but not limited to those isolated from one or more representatives of the *Streptomyces* genus. The Actinobacterium of the *Streptomyces* genus include, but are not limited to, *Streptomyces abietis*, *Streptomyces abikoensis*, *Streptomyces aburaviensis*, *Streptomyces achromogenes*, *Streptomyces acidiscabies*, *Streptomyces actinomycinicus*, *Streptomyces acrimycinii*, *Streptomyces actuosus*, *Streptomyces aculeolatus*, *Streptomyces abyssalis*, *Streptomyces afghaniensis*, *Streptomyces ailingensis*, *Streptomyces africanus*, *Streptomyces alanosinicus*, *Streptomyces albaduncus*, *Streptomyces albiaxialis*, *Streptomyces albidochromogenes*, *Streptomyces albiflavescens*, *Streptomyces albiflaviniger*, *Streptomyces albidoflavus*, *Streptomyces albofaciens*, *Streptomyces alboflavus*, *Streptomyces albogriseolus*, *Streptomyces albolongus*, *Streptomyces alboniger*, *Streptomyces albospinus*, *Streptomyces albulus*, *Streptomyces albus*, *Streptomyces aldersoniae*, *Streptomyces alfalfa*, *Streptomyces alkaliphilus*, *Streptomyces alkali-thermotolerans*, *Streptomyces almquistii*, *Streptomyces alni*, *Streptomyces althioticus*, *Streptomyces amakusaensis*, *Streptomyces ambofaciens*, *Streptomyces amritsarensis*, *Streptomyces anandii*, *Streptomyces angustmyceticus*, *Streptomyces anthocyanicus*, *Streptomyces antibioticus*, *Streptomyces antimycoticus*, *Streptomyces anulatus*, *Streptomyces aomiensis*, *Streptomyces araujonae*, *Streptomyces ardens*, *Streptomyces arenae*, *Streptomyces armeniacus*, *Streptomyces artemisiae*, *Streptomyces arcticus*, *Streptomyces ascomycinicus*, *Streptomyces asiaticus*, *Streptomyces asterosporus*, *Streptomyces atacamensis*, *Streptomyces atratus*, *Streptomyces atriruber*, *Streptomyces atroolivaceus*, *Streptomyces atrovirens*, *Streptomyces aurantiacus*, *Streptomyces aurantiogriseus*, *Streptomyces auratus*, *Streptomyces aureocirculatus*, *Streptomyces aureofaciens*, *Streptomyces aureo-rectus*, *Streptomyces aureovorticellatus*, *Streptomyces aureus*, *Streptomyces avellaneus*, *Streptomyces avermitilis*, *Streptomyces avicenniae*, *Streptomyces avidinii*, *Streptomyces axinellae*, *Streptomyces azureus*, *Streptomyces bacillaris*, *Streptomyces badius*, *Streptomyces bambergensis*, *Streptomyces bangladeshensis*, *Streptomyces baliensis*, *Streptomyces barkulensis*, *Streptomyces beijiangensis*, *Streptomyces bellus*, *Streptomyces bikiniensis*, *Streptomyces*

blastmyceticus, *Streptomyces bluensis*, *Streptomyces bobili*, *Streptomyces bohaisensis*, *Streptomyces bottropensis*, *Streptomyces brasiliensis*, *Streptomyces brevispora*, *Streptomyces bullii*, *Streptomyces bungoensis*, *Streptomyces burgazadensis*, *Streptomyces cacaoi*, *Streptomyces caelestis*, *Streptomyces caeruleatus*, *Streptomyces calidiresistens*, *Streptomyces calvus*, *Streptomyces canaries*, *Streptomyces canchipurensis*, *Streptomyces candidus*, *Streptomyces cangkringensis*, *Streptomyces caniferus*, *Streptomyces canus*, *Streptomyces capillispiralis*, *Streptomyces capoaensis*, *Streptomyces carpaticus*, *Streptomyces carpinensis*, *Streptomyces castelarensis*, *Streptomyces catbensis*, *Streptomyces catenulae*, *Streptomyces cavourensis*, *Streptomyces cello staticus*, *Streptomyces celluloflavus*, *Streptomyces cellulolyticus*, *Streptomyces cellulosa*, *Streptomyces chartreusis*, *Streptomyces chattanoogensis*, *Streptomyces cheonanensis*, *Streptomyces Chiangmaiensis*, *Streptomyces chrestomyceticus*, *Streptomyces chromofuscus*, *Streptomyces chryseus*, *Streptomyces chilikensis*, *Streptomyces chlorus*, *Streptomyces chumphonensis*, *Streptomyces cinereorectus*, *Streptomyces cinereoruber*, *Streptomyces cinereospinus*, *Streptomyces cinereus*, *Streptomyces cinerochromogenes*, *Streptomyces cinnabarinus*, *Streptomyces cinnamomensis*, *Streptomyces cinnamoneus*, *Streptomyces cirratus*, *Streptomyces ciscaucasicus*, *Streptomyces clavifer*, *Streptomyces clavuligerus*, *Streptomyces coacervatus*, *Streptomyces cocklensis*, *Streptomyces coelestis*, *Streptomyces coelicoflavus*, *Streptomyces coelicolor*, *Streptomyces coeruleoflavus*, *Streptomyces coeruleofuscus*, *Streptomyces coeruleoprurus*, *Streptomyces coeruleorubidus*, *Streptomyces coeruleus*, *Streptomyces collinus*, *Streptomyces colombiensis*, *Streptomyces corchorusii*, *Streptomyces costaricanus*, *Streptomyces cremeus*, *Streptomyces crystallinus*, *Streptomyces cupidosporus*, *Streptomyces cyaneofuscatus*, *Streptomyces cyaneus*, *Streptomyces cyanoalbus*, *Streptomyces cyslabdanicus*, *Streptomyces daghestanicus*, *Streptomyces daliensis*, *Streptomyces daqingensis*, *Streptomyces deccanensis*, *Streptomyces decoyicus*, *Streptomyces demainii*, *Streptomyces deserti*, *Streptomyces diastaticus*, *Streptomyces diastatochromogenes*, *Streptomyces djakartensis*, *Streptomyces drozdowiczii*, *Streptomyces durhamensis*, *Streptomyces durmitorensis*, *Streptomyces echinatus*, *Streptomyces echinoruber*, *Streptomyces ederensis*, *Streptomyces emeiensis*, *Streptomyces endophyticus*, *Streptomyces endus*, *Streptomyces enissocaesilis*, *Streptomyces erythraeus* (also known as *Saccharopolyspora erythraea*), *Streptomyces erythrogriseus*, *Streptomyces erringtonii*, *Streptomyces eurocidicus*, *Streptomyces europaeiscabiei*, *Streptomyces eurythermus*, *Streptomyces exfoliates*, *Streptomyces faba*, *Streptomyces fenghuangensis*, *Streptomyces ferralitidis*, *Streptomyces filamentosus*, *Streptomyces fildesensis*, *Streptomyces filipinensis*, *Streptomyces fimbriatus*, *Streptomyces finlayi*, *Streptomyces flaveolus*, *Streptomyces flaveus*, *Streptomyces flavofungini*, *Streptomyces flavotricini*, *Streptomyces flavovariabilis*, *Streptomyces flavovirens*, *Streptomyces flavoviridis*, *Streptomyces fradiae*, *Streptomyces fragilis*, *Streptomyces fukangensis*, *Streptomyces fulvissimus*, *Streptomyces fulvorubeus*, *Streptomyces fumanus*, *Streptomyces fumigatiscleroticus*, *Streptomyces galbus*, *Streptomyces galilaeus*, *Streptomyces gancidicus*, *Streptomyces gardneri*, *Streptomyces gelaticus*, *Streptomyces geldanamycininus*, *Streptomyces geysiriensis*, *Streptomyces ghanaensis*, *Streptomyces gilvifuscus*, *Streptomyces glaucescens*, *Streptomyces glauciniger*, *Streptomyces glaucosporus*, *Streptomyces glaucus*, *Streptomyces globisporus*, *Streptomyces globosus*, *Streptomyces glomeratus*, *Streptomyces glomeroaurantiacus*, *Streptomyces glycovorans*, *Streptomyces*

ces gobitricini, Streptomyces goshikiensis, Streptomyces gougerotii, Streptomyces gramineus, Streptomyces gramineus, Streptomyces graminifolii, Streptomyces graminilatus, Streptomyces graminisoli, Streptomyces gri-
seiniger, Streptomyces griseoaurantiacus, Streptomyces
griseocarneus, Streptomyces griseochromogenes, Strepto-
myces griseoflavus, Streptomyces griseofuscus, Streptomy-
ces griseoincarnatus, Streptomyces griseoloalbus, Strepto-
myces griseolus, Streptomyces griseoluteus, Streptomyces
griseomycini, Streptomyces griseoplanus, Streptomyces
griseorubens, Streptomyces griseoruber, Streptomyces
griseorubiginosus, Streptomyces griseosporeus, Streptomy-
ces griseostramineus, Streptomyces griseoviridis, Strepto-
myces griseus, Streptomyces guanduensis, Streptomyces
gulbargensis, Streptomyces hainanensis, Streptomyces hali-
clonae, Streptomyces halophytocola, Streptomyces halstedii,
Streptomyces harbinensis, Streptomyces hawaiiensis, Strepto-
myces hebeiensis, Streptomyces heilongjiangensis, Strepto-
myces heliomycini, Streptomyces helveticus, Streptomyces
herbaceous, Streptomyces herbaricolor, Streptomyces
himastatinicus, Streptomyces hirosheimensis, Streptomyces
hirsutus, Streptomyces hokutonensis, Streptomyces hoynat-
ensis, Streptomyces humidus, Streptomyces humiferus,
Streptomyces hundungensis, Streptomyces hyderabadensis,
Streptomyces hygrosopicus, Streptomyces hypolithicus,
Streptomyces iakyrus, Streptomyces iconiensis, Streptomy-
ces incanus, Streptomyces indiaensis, Streptomyces indigo-
ferus, Streptomyces indicus, Streptomyces indonesiensis,
Streptomyces intermedius, Streptomyces inusitatus, Strepto-
myces ipomoeae, Streptomyces iranensis, Streptomyces jan-
thinus, Streptomyces javensis, Streptomyces jietaisiensis,
Streptomyces jiujiangensis, Streptomyces kaempferi, Strepto-
myces kanamyceticus, Streptomyces karpasiensis, Strepto-
myces kasugaensis, Streptomyces katrae, Streptomyces
kebangsaanensis, Streptomyces klenkii, Streptomyces koy-
angensis, Streptomyces kunmingensis, Streptomyces kurs-
sanovii, Streptomyces labedae, Streptomyces lacrimiflumi-
nis, Streptomyces lacticiproducens, Streptomyces
laculatispora, Streptomyces lanatus, Streptomyces lannen-
sis, Streptomyces lateritius, Streptomyces laurentii, Strepto-
myces lavendofoliae, Streptomyces lavendulae, Streptomy-
ces lavenduligriseus, Streptomyces leeuwenhoekii,
Streptomyces lavendulicolor, Streptomyces levis, Strepto-
myces libani, Streptomyces lienomycini, Streptomyces lilaci-
nus, Streptomyces lincolnensis, Streptomyces litmoxidini,
Streptomyces litalalis, Streptomyces lomondensis, Strepto-
myces longisporoflavus, Streptomyces longispororuber,
Streptomyces lopnurensis, Streptomyces longisporus, Strepto-
myces longwoodensis, Streptomyces lucensis, Streptomy-
ces lunaelactis, Streptomyces lunalinharesii, Streptomyces
luridiscabiei, Streptomyces luridus, Streptomyces lusitanus,
Streptomyces lushanensis, Streptomyces luteireticuli, Strepto-
myces luteogriseus, Streptomyces luteosporeus, Strepto-
myces lydicus, Streptomyces macrosporus, Streptomyces
malachitofuscus, Streptomyces malachitospinus, Streptomy-
ces malaysiensis, Streptomyces mangrove, Streptomyces
marinus, Streptomyces marokkonensis, Streptomyces
masuensis, Streptomyces massasporeus, Streptomyces mat-
ensis, Streptomyces mayteni, Streptomyces mauvecolor,
Streptomyces megaspores, Streptomyces melanogenes,
Streptomyces melanosporefaciens, Streptomyces mexicanus,
Streptomyces michiganensis, Streptomyces microflavus,
Streptomyces milbemycinicus, Streptomyces minutiscleroti-
cus, Streptomyces mirabilis, Streptomyces misakiensis,
Streptomyces misionensis, Streptomyces mobaraensis,
Streptomyces monomycini, Streptomyces mordarskii, Strepto-
myces morookaense, Streptomyces muensis, Streptomyces

murinus, Streptomyces mutabilis, Streptomyces mutomycini,
Streptomyces naganishii, Streptomyces nanhaiensis, Strepto-
myces nanshensis, Streptomyces narbonensis, Streptomy-
ces nashvillensis, Streptomyces netropsis, Streptomyces ney-
agawaensis, Streptomyces niger, Streptomyces nigrescens,
Streptomyces nitrosporeus, Streptomyces niveiscabiei,
Streptomyces niveiscabiei, Streptomyces niveoruber, Strepto-
myces niveus, Streptomyces noboritoensis, Streptomyces
nodosus, Streptomyces nogalater, Streptomyces nojiriensis,
Streptomyces noursei, Streptomyces novaecaesareae, Strepto-
myces ochraceiscleroticus, Streptomyces olivaceisclerotici-
cus, Streptomyces olivaceoviridis, Streptomyces olivaceus,
Streptomyces olivicoloratus, Streptomyces olivochromo-
genes, Streptomyces olivomycini, Streptomyces olivovorticil-
latus, Streptomyces omiyaensis, Streptomyces osmaniensis,
Streptomyces orinoci, Streptomyces pactum, Streptomyces
panacagri, Streptomyces panaciradicis, Streptomyces para-
doxus, Streptomyces parvulus, Streptomyces parvus, Strepto-
myces pathocidini, Streptomyces paucisporeus, Strepto-
myces peucetius, Streptomyces phaeochromogenes,
Streptomyces phaeofaciens, Streptomyces phaeogriseichromato-
genes, Streptomyces phaeoluteichromatogenes, Strepto-
myces phaeoluteigriseus, Streptomyces phaeopurpureus,
Streptomyces pharetrae, Streptomyces pharamamarensis,
Streptomyces phytohabitans, Streptomyces pilosus, Strepto-
myces platensis, Streptomyces plicatus, Streptomyces plum-
biresistens, Streptomyces pluricologrescens, Streptomyces
pluripotens, Streptomyces polyanthibioticus, Streptomyces
polychromogenes, Streptomyces polygonati, Streptomyces
polymachus, Streptomyces poonensis, Streptomyces pra-
sinopilosus, Streptomyces prasinosporus, Streptomyces
prasinus, Streptomyces pratens, Streptomyces pratensis,
Streptomyces prunicolor, Streptomyces psammoticus, Strepto-
myces pseudoechinosporus, Streptomyces pseudogriseo-
lus, Streptomyces pseudovenezuelae, Streptomyces pul-
veraceus, Streptomyces puniceus, Streptomyces
puniciscabiei, Streptomyces purpeofuscus, Streptomyces
purpurascens, Streptomyces purpureus, Streptomyces pur-
purogeniscleroticus, Streptomyces qinglanensis, Strepto-
myces racemochromogenes, Streptomyces radiopugnans,
Streptomyces rameus, Streptomyces ramulosus, Streptomy-
ces rapamycinicus, Streptomyces recifensis, Streptomyces
rectiviolaceus, Streptomyces regensis, Streptomyces resisto-
mycificus, Streptomyces reticuliscabiei, Streptomyces rhizo-
philus, Streptomyces rhizosphaericus, Streptomyces rimo-
sus, Streptomyces rishiriensis, Streptomyces rochei,
Streptomyces rosealbus, Streptomyces roseiscleroticus,
Streptomyces roseofulvus, Streptomyces roseolilacinus,
Streptomyces roseolus, Streptomyces roseosporus, Strepto-
myces roseoviolaceus, Streptomyces roseoviridis, Strepto-
myces ruber, Streptomyces rubidus, Streptomyces rubigino-
sohelvolus, Streptomyces rubiginosus, Streptomyces
rubrisoli, Streptomyces rubrogriseus, Streptomyces rubrus,
Streptomyces rutgersensis, Streptomyces samsunensis,
Streptomyces sanglieri, Streptomyces sannanensis, Strepto-
myces sanyensis, Streptomyces sasae, Streptomyces scabiei,
Streptomyces scabrisporus, Streptomyces sclerotialis,
Streptomyces scopiformis, Streptomyces scopuliridis, Strepto-
myces sedi, Streptomyces seoulensis, Streptomyces serani-
matus, Streptomyces seymenliensis, Streptomyces shaanxi-
ensis, Streptomyces shenzhenensis, Streptomyces
showdoensis, Streptomyces silaceus, Streptomyces sinden-
ensis, Streptomyces sioyaensis, Streptomyces smyrnaeus,
Streptomyces sodiophilus, Streptomyces somaliensis, Strepto-
myces sudanensis, Streptomyces sparsogenes, Streptomy-
ces sparsus, Streptomyces specialis, Streptomyces spectabi-
lis, Streptomyces speibonae, Streptomyces speleomycini,

Streptomyces spinoverrucosus, *Streptomyces spiralis*, *Streptomyces spiroverticillatus*, *Streptomyces spongiae*, *Streptomyces spongiicola*, *Streptomyces sporocinereus*, *Streptomyces sporoclivatus*, *Streptomyces spororaveus*, *Streptomyces sporoverrucosus*, *Streptomyces staurosporininus*, *Streptomyces stelliscabiei*, *Streptomyces stramineus*, *Streptomyces subrutilis*, *Streptomyces sulfonofaciens*, *Streptomyces sulphurous*, *Streptomyces sundarbansensis*, *Streptomyces synnematoformans*, *Streptomyces tacrolimicus*, *Streptomyces tanashiensis*, *Streptomyces tateyamensis*, *Streptomyces tauricus*, *Streptomyces tendae*, *Streptomyces termitum*, *Streptomyces thermoalcalitolerans*, *Streptomyces thermoautotrophicus*, *Streptomyces thermocarboxydovorans*, *Streptomyces thermocarboxyidus*, *Streptomyces thermocoprophilus*, *Streptomyces thermodiastaticus*, *Streptomyces thermogriseus*, *Streptomyces thermolineatus*, *Streptomyces thermospinosporus*, *Streptomyces thermoviolaceus*, *Streptomyces thermovulgaris*, *Streptomyces thinghirensis*, *Streptomyces thioluteus*, *Streptomyces torulosus*, *Streptomyces toxystriatus*, *Streptomyces tremellae*, *Streptomyces tritolerans*, *Streptomyces tricolor*, *Streptomyces tsukubensis*, *Streptomyces tubercidicus*, *Streptomyces tuius*, *Streptomyces tunisiensis*, *Streptomyces turgidiscabies*, *Streptomyces tyrosinilyticus*, *Streptomyces umbrinus*, *Streptomyces variabilis*, *Streptomyces variegatus*, *Streptomyces varsoviensis*, *Streptomyces verticillus*, *Streptomyces vastus*, *Streptomyces venezuelae*, *Streptomyces vietnamensis*, *Streptomyces vinaceus*, *Streptomyces vinaceusdrappus*, *Streptomyces violaceochromogenes*, *Streptomyces violaceolatus*, *Streptomyces violaceorectus*, *Streptomyces violaceoruber*, *Streptomyces violaceorubidus*, *Streptomyces violaceus*, *Streptomyces violaceusniger*, *Streptomyces violarius*, *Streptomyces violascens*, *Streptomyces violens*, *Streptomyces virens*, *Streptomyces virginiae*, *Streptomyces viridis*, *Streptomyces viridiviolaceus*, *Streptomyces viridobrunneus*, *Streptomyces viridochromogenes*, *Streptomyces viridodiastaticus*, *Streptomyces viridosporus*, *Streptomyces vitaminophilus*, *Streptomyces wedmorensis*, *Streptomyces wellingtoniae*, *Streptomyces werraensis*, *Streptomyces wuyuanensis*, *Streptomyces xanthochromogenes*, *Streptomyces xanthocidicus*, *Streptomyces xantholiticus*, *Streptomyces xanthophaeus*, *Streptomyces xiamenensis*, *Streptomyces xinghaiensis*, *Streptomyces xishensis*, *Streptomyces yaanensis*, *Streptomyces yanglinensis*, *Streptomyces yangpuensis*, *Streptomyces yanii*, *Streptomyces yatensis*, *Streptomyces yeochonensis*, *Streptomyces yerevanensis*, *Streptomyces yogyakartansis*, *Streptomyces yokosukanensis*, *Streptomyces youssoufiensis*, *Streptomyces yunnanensis*, *Streptomyces zagrosensis*, *Streptomyces zaomyceticus*, *Streptomyces zhaozhouensis*, *Streptomyces zinciresistens*, or *Streptomyces ziwulingensis*.

As used herein, the term “silent” or “quiescent”, when used in reference to a gene, refers to a gene that has no phenotypical effect on the host and/or has no detectable expression. This non-effect of a silent gene can be due to the either low or non-existent expression of the silent gene. The term “silent gene” may also refer to a transcriptionally inactive gene. As used herein, “silent gene” refers to a gene that is unable to express the associated gene product from its coding sequence, either during transcription or translation processes in the cellular host.

As used herein, the term “activation” refers to an upregulation of gene expression or transcriptional activation of a gene that was previously not expressed or only expressed in small amounts. Conversely, the term “suppression” or “repression” refers to a downregulation of gene expression or transcriptional activity of a gene.

As used herein, the term “co-linear” refers to open reading frames that are transcribed in the same direction.

As used herein, the term “trans-conjugation” or “conjugation” refers to the transfer of genetic material between bacterial cells by horizontal gene transfer, e.g., by direct cell-to-cell contact or by a bridge-like connection between two cells.

As used herein, the term “antimicrobial” includes antibiotics and chemicals capable of inhibiting or preventing the growth of, or capable of killing, microbes, especially bacteria. An example of an antimicrobial chemical is a disinfectant. Classes of antimicrobials are known in the art. See, e.g., Vittorio Tracanna, Anne de Jong, Marnix H. Medema, Oscar P. Kuipers; Mining prokaryotes for antimicrobial compounds: from diversity to function, FEMS Microbiology Reviews, Volume 41, Issue 3, 1 May 2017, Pages 417-429, at Table 1, incorporated herein by reference.

As used herein, the term “antibiotic” refers to an agent produced by a living organism (e.g., a bacterium) that is capable of inhibiting the growth of another living organism (e.g., another bacterium) or that is capable of killing another living organism (e.g., another bacterium).

As used herein, the term “artificial DNA” refers to a DNA molecule or a portion of a DNA molecule that is different from any found in nature or that is produced by non-natural processes, for example, as the result of in vitro techniques or solid-phase DNA synthesis.

As used herein, the term “isolated DNA sequence” refers to any DNA molecule, however constructed or synthesized, that is locationally distinct from its natural location in genomic and/or episomal DNA. The definition includes the isolated DNA sequence in all its forms other than the natural state. For example, the DNA sequence may be inserted into a plasmid or phage vector or inserted into the genome of the organism from which it came or any other organism.

As used herein, the term “gene” refers to a polynucleotide comprising coding sequence for at least one polypeptide or non-coding RNA. When referring to protein-coding genes, the term “gene” refers to a polynucleotide comprising at least one open reading frame that is capable of encoding a particular polypeptide or protein after being transcribed and translated. Any of the polynucleotide sequences described herein may be used to identify larger fragments or full-length coding sequences of the gene with which they are associated.

As used herein, the term “synthetic gene” refers to a DNA fragment synthesized in the laboratory by combining nucleotides without preexisting DNA sequences. In particular, the term refers to a completely synthetic double-stranded DNA molecule.

As used herein, the term “recognition site” refers to a location on a DNA molecule containing specific sequences of nucleotides that are recognized by specific proteins or enzymes or by specific nucleic acids.

As used herein, the term “restriction endonuclease” refers to an enzyme that cuts a double-stranded DNA molecule at a specific recognition site. In some embodiments of the present technology, the term relates to restriction endonucleases or enzymes that specifically recognize DNA sequences of 6, 7, or 8 nucleotides, in which the nucleotide sequence of one DNA strand reads in reverse order to that of the complementary DNA strand (palindromic). However, the technology is not limited to use of such enzymes and includes restriction endonucleases that have recognition sequences of other sizes.

As used herein, the term oriV refers to the origin of replication (e.g., derived from the bacterial F plasmid) that

finds use in some embodiments of the expression vector as described herein; ori2 refers to the secondary origin of replication, which is also known as oriS (e.g., derived from the bacterial F plasmid) that finds use in some embodiments of the expression vector described herein; repE refers to a gene encoding the replication initiation protein (e.g., derived from the bacterial F plasmid) that finds use in some embodiments of the expression vector described herein; incC refers to the incompatibility region (e.g., derived from the bacterial F plasmid) that finds use in some embodiments of the expression vector described herein; sopA refers to a gene encoding a partitioning protein (e.g., derived from the bacterial F plasmid) that finds use in some embodiments of the expression vector described herein; sopB refers to a gene encoding a partitioning protein (e.g., derived from the bacterial F plasmid) that finds use in some embodiments of the expression vector described herein; sopC refers to a gene encoding a partitioning protein (e.g., derived from the bacterial F plasmid) that finds use in some embodiments of the expression vector described herein; oriT refers to the incP origin of transfer for some embodiments of the expression vector described herein; ApramR refers to the aac (3)-IV apramycin resistance gene that finds use in some embodiments of the expression vector described herein; the phage ϕ C31 attP site allows integration to genomic attB sites and is a component that finds use in some embodiments of the expression vector described herein; the phage ϕ C31 integrase allows integration between attP and attB sites and is a component that finds use in some embodiments of the expression vector described herein; cos refers to the lambda cos site, which allows packaging into phage lambda particles, that finds use in some embodiments of the expression vector described herein; and Kanamycin-r refers to a gene encoding the kanamycin resistance gene that finds use in some embodiments of the expression vector described herein.

As used herein, the term “Cas9” (CRISPR associated protein 9) refers to an RNA-guided DNA endonuclease enzyme associated with the CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) adaptive immunity system in *Streptococcus pyogenes*, among other bacteria. Cas9 can be used to cleave DNA in vitro or in vivo by use of sequence specific guide RNA to target a known region of DNA.

As used herein, the term “target DNA” or “target vector” refers to a double-stranded DNA that is suitable to be modified using molecular biology techniques. In this technology, the definition relates to episomal DNA sequences that include specific recognition sites for restriction endonucleases or that can be modified by transposases as a result of the inclusion of transposable DNA elements.

As used herein, the term “origin of replication” or “replication origin” refers to particular sequences in episomal DNAs at which replication is initiated, based on recruiting proteins involved in DNA replication.

As used herein, the term “selectable marker” refers to a gene whose expression allows one to identify cells that have been transformed or transfected with a vector containing the marker gene (e.g., by antibiotic resistance on antibiotic medium, fluorescence, color generation, or other detectable signal). For instance, a recombinant nucleic acid may include a selectable marker operably linked to a gene of interest and a promoter, such that expression of the selectable marker indicates the successful transformation of the cell with the gene of interest. In some embodiments, a “selectable marker” refers to a gene located inside bacteria (at genomic or episomal level) that confers a feature for

artificial selection. The term is typically associated with antibiotic resistance genes (e.g., a chloramphenicol resistance gene) provided in vectors or artificial vectors for selection of bacterial isolates after transformation.

As used herein, the term “transformation” refers to the process of introducing genetic material into a cell, e.g., to bacterial cells. In some embodiments of the present technology, the term is associated with introducing vectors, expression vectors, modified artificial vectors, and clones (e.g., comprising a vector and an insert) into bacterial cells.

As used herein, the terms “upstream” and “downstream” refer to relative positions of portions of nucleic acids or nucleic acid sequences (e.g., DNA or RNA) and are often used to differentiate relative positions in DNA or RNA sequences. As used herein, the terms “upstream” and “downstream” are defined relative to the 5' to 3' direction in which RNA transcription takes place. “Upstream” is toward the 5' end of the RNA molecule and “downstream” is toward the 3' end. For double-stranded DNA, “upstream” is toward the 5' end of the coding strand for the nucleic acid and downstream is toward the 3' end. Thus, in exemplary use, “upstream” is a position towards the 5' from another nucleic acid segment (e.g., promoter, gene, restriction site, etc.) in a single strand of DNA or in a RNA molecule and “downstream” is a position towards the 3' from another nucleic acid segment in a single strand of DNA or in a RNA molecule.

As used herein, the term “metagenome” is defined as “the collective genomes of all microorganisms present in a given habitat” (Handelsman et al., (1998) Chem. Biol. 5: R245-R249). However, this term is also intended to include clones, including the genomes or genes extracted from environmental samples.

As used herein, “metagenomic DNA” refers to the whole microbial-associated genomic DNA isolated from complex samples like open natural environments (e.g. soil, water) or from microbiomes of multicellular organisms (e.g. humans).

As used herein, “metagenomic library” refers to a clone collection of whole microbial-associated genomic DNA isolated from complex samples like open natural environments (e.g. soil, water) or from microbiomes of multicellular organisms (e.g. humans) in a recombinant vector.

As used herein, “genome” refers to the genetic material (e.g., chromosome) of an organism.

As used herein, “PCR” refers to the polymerase chain reaction method of amplifying DNA or RNA.

As used herein, “antiSMASH” refers to a software program used to identify motifs commonly found in BGCs.

As used herein, “insert” or “DNA insert” refers to a piece or fragment or sequence of DNA that is inserted, by molecular biology techniques, into a vector or an artificial vector for its subsequent selection, propagation, manipulation, or expression in a host organism.

As used herein, the term “large plasmid” or “large vector” refers to a plasmid or vector (e.g., an expression vector) that is larger than 10 kb and/or that comprises an insert greater than 5-10 kb. Large inserts (e.g., greater than 5-10 kb) are unstable (e.g., through recombination) at high copy numbers. In some embodiments, partitioning factors (e.g., sopA, sopB, and/or sopC) help maintain a plasmid or vector at single copy number to minimize and/or eliminate recombination instability.

As used herein, “reporter gene” means a gene whose expression in a bacterial host can be easily monitored or detected. In some embodiments of the present technology, the reporter gene encodes for a green fluorescent protein (GFP) variant.

As used herein, the term “gene” refers to a nucleic acid molecule that comprises a nucleic acid sequence that encodes a polypeptide or non-coding RNA and the expression control sequences that are operably linked to the nucleic acid sequence that encodes the polypeptide or non-coding RNA. For instance, a gene may comprise a promoter, one or more enhancers, a nucleic acid sequence that encodes a polypeptide or a non-coding RNA, downstream regulatory sequences and, possibly, other nucleic acid sequences involved in regulating the transcription of an RNA from the gene.

A nucleic acid molecule or polypeptide is “derived” from a particular species if the nucleic acid molecule or polypeptide has been isolated from the particular species or if the nucleic acid molecule or polypeptide is homologous to a nucleic acid molecule or polypeptide isolated from a particular species.

As used herein, “kilobase” (kb) or “kilobase pairs” (kbp) refers to 1000 nucleotides or 1000 base pairs of a nucleic acid (e.g., DNA or RNA).

As used herein, the term “in vitro” refers to studies that are conducted using components of an organism that have been isolated from their usual biological surroundings to permit a more detailed or more convenient analysis than can be done with whole organisms. Colloquially, these experiments are commonly called “test tube experiments”. In contrast, in vivo studies are those that are conducted with living organisms in their normal intact state.

A used herein, the term “in vivo” refers to experimentation using living cells or a whole living organism as opposed to a partial or dead cell or organism, or an in vitro (“within the glass”, e.g., in a test tube or petri dish) controlled environment.

As used herein, the term “molecular cloning” refers to the method of preparing or assembling exogenous, homologous, and/or heterologous DNA for propagation, selection, and/or expression within a host organism. In a conventional molecular cloning experiment, the DNA to be cloned is obtained from an organism or metagenome of interest and subsequently treated with enzymes such as Cas9 or restriction enzymes in to generate smaller DNA fragments. Subsequently, these fragments are then combined with vector DNA to produce recombinant DNA molecules. The recombinant DNA is then introduced into a host organism (typically an easy-to-grow, benign, laboratory strain of *E. coli* bacteria) to produce a population of organisms in which recombinant DNA molecules are replicated along with the host DNA. This process takes advantage of the fact that a single bacterial cell can be induced to take up and replicate a single recombinant DNA molecule. This single cell can then be expanded exponentially to generate a large amount of bacteria, each of which contains a copy of the original recombinant molecule. Thus, both the resulting bacterial population, and the recombinant DNA molecule, are commonly referred to as “clones”.

The method of molecular cloning can also be used to regulate gene expression. In general, regulation of gene expression comprises and includes a wide range of mechanisms that are used by cells to increase or decrease the production of specific gene products (protein or RNA). Sophisticated programs of gene expression are widely observed and known in the art, for example, as a mechanism to trigger developmental pathways, respond to environmental stimuli, or adapt to new food sources. Virtually any step of gene expression can be modulated, from transcriptional initiation, to RNA processing, and to the post-translational modification of a protein. Often, one gene regulator controls

another, and so on, resulting in a complex gene regulatory network. The process of gene expression itself can be divided into two major processes, transcription and translation. One place in which the production of specific gene products can be influenced is during transcription, which is the process of transcribing DNA to RNA, which ultimately has an effect on the protein expressed during a later process called translation (also known as protein synthesis). Transcriptional regulation is the means by which a cell regulates the conversion of DNA to RNA (transcription), thereby orchestrating gene activity. A single gene can be regulated in a range of ways, from altering the number of copies of RNA that are transcribed to the temporal control of when the gene is transcribed. Transcriptional regulation also influences temporal expression of particular proteins. This control allows the cell or organism to respond to a variety of intra- and extracellular signals and thus mount a response. Some examples of this include producing the mRNA that encode enzymes to adapt to a change in a food source, producing the gene products, including proteins, involved in cell cycle specific activities, and producing the gene products, including proteins, responsible for cellular differentiation in higher eukaryote.

Percentage identity determinations can be performed for nucleic acids using BLASTN or standard nucleotide BLAST using default settings (Match/Mismatch scores 1, -2) Gap costs linear, Expect threshold 10, Word size 28, and match matches in a query range 0) and for proteins using BLAST using default settings (Expect threshold 10, Word size 3, Max matches in a query range 0, Matrix Blosum62, Gap costs Existence 11, extension 1 and conditional compositional score matrix adjustment).

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features, aspects, and advantages of the present technology will become better understood with regard to the following drawings.

In the drawings, oriV refers to the origin of replication for the bacterial F plasmid; ori2 refers to the secondary origin of replication for the bacterial F plasmid; also known as oriS; repE refers to a gene encoding the replication initiation protein for the bacterial F plasmid; incC refers to the incompatibility region of the bacterial F plasmid; sopA refers to a gene encoding a partitioning protein for the bacterial F plasmid; sopB refers to a gene encoding a partitioning protein for the bacterial F plasmid; sopC refers to a gene encoding a partitioning protein for the bacterial F plasmid; oriT refers to the incP origin of transfer; ApramR refers to the aac (3)-IV apramycin resistance gene; the phage ϕ C31 attP site allows integration to genomic attB sites; the phage ϕ C31 integrase allows integration between attP and attB sites; cos refers to the lambda cos site, which allows packaging into phage lambda particles; and Kanamycin-r refers to a gene encoding the kanamycin resistance gene.

FIG. 1 shows three embodiments of the dual promoter cassette comprising OTC-promoter Potr and ϵ -cap promoter PnitA and the indicator GFP (green fluorescent protein) gene used in the BAC expression vector described herein. Variants 1 (dualP1) and 2 (dualP2) comprise GFP in an orientation for OTC induction while variant 3 (dualP3) comprises GFP in an orientation for ϵ -cap induction. Variants 2 and 3 comprise PacI restriction sites, which are rare and useful for cloning and subsequent modifications. OtrR encodes a regulator for Potr. NitR encodes a regulator for PnitA.

FIG. 2 shows sfGFP induction and fluorescence in *E. coli*. Presence of the inducers allows for expression of sfGFP and detection by UV fluorescence.

FIG. 3 shows a plasmid map of an embodiment of the dual inducible promoter expression vector described herein ("pDualP"). This expression vector is useful for cloning DNA to *E. coli* and conjugation to, genome integration in, and inducible expression in *Streptomyces* and other organisms. Nucleotide sequences of Potr, OtrR, sfGFP, NitR, PnitA, and the dual promoter sequence comprising the sfGFP insert are provided by SEQ ID NOs: 1, 3, 4, 5, 6, and 7, respectively.

FIG. 4 is a schematic showing a method for subcloning metagenomic BGCs. Metagenomic BGCs from a BAC library clone are restricted by Cas9 at two unique sites flanking the BGC and assembled into a pDualP expression vector containing overlaps that match the ends of the restricted BGC.

FIG. 5 shows ACT and RED BGCs cloned to pDualP in both orientations. Model *S. coelicolor* BGCs ACT and RED were cloned to pDualP in both orientations.

FIG. 6 shows pDualP ACT induction on MS agar imaged from either the front of the plate or the back of the plate, through the agar. In *S. lividans* Δ act Δ red, the ACT BGC in both cloning orientations in pDualP is activated in response to inducers while the ACT BGC cloned without promoters is not activated.

FIG. 7 shows pDualP RED induction on MS agar. In *S. lividans* Δ act Δ red, the RED BGC in both cloning orientations in pDualP is activated in response to inducers while the RED BGC cloned without promoters is not activated.

FIG. 8 shows pDualP induction of ACT in YEME broth or RED in R2YE liquid broth. In *S. lividans* Δ act Δ red, the ACT and RED BGCs cloned to pDualP are activated in response to inducers while ACT and RED BGCs cloned without promoters are not activated.

FIG. 9 shows a quantitative analysis of the pDualP inducible expression system in *S. coelicolor* M1154. Two metagenomic-derived BGCs were cloned to pDualP, introduced to *S. coelicolor* M1154, and extracts tested in an antibiosis activity assay against *A. baumannii*. Both BGCs show increased antibiosis activity in response to ϵ -cap. The growth inhibition of *A. baumannii* 3806 by supernatants of *S. coelicolor* clones harboring metagenomic BGCs with and without inducible expression is depicted. Inducible expression by pDualP is compared to the expression of the *S. coelicolor* native promoter (black bar). The values presented are the percent inhibitions of *A. baumannii* 3806 relative to the inhibition by the empty expression vector control from three replicates $\pm$ SD of each treatment group.

It is to be understood that the figures are not necessarily drawn to scale, nor are the objects in the figures necessarily drawn to scale in relationship to one another. The figures are depictions that are intended to bring clarity and understanding to various embodiments of apparatuses, systems, and methods disclosed herein. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. Moreover, it should be appreciated that the drawings are not intended to limit the scope of the present teachings in any way.

DETAILED DESCRIPTION

In this detailed description of the various embodiments, for purposes of explanation, numerous specific details are set forth to provide a thorough understanding of the embodiments disclosed. One skilled in the art will appreciate,

however, that these various embodiments may be practiced with or without these specific details. In other instances, structures and devices are shown in block diagram form. Furthermore, one skilled in the art can readily appreciate that the specific sequences in which methods are presented and performed are illustrative and it is contemplated that the sequences can be varied and still remain within the spirit and scope of the various embodiments disclosed herein.

All literature and similar materials cited in this application, including but not limited to, patents, patent applications, articles, books, treatises, and internet web pages are expressly incorporated by reference in their entirety for any purpose. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of ordinary skill in the art to which the various embodiments described herein belongs. When definitions of terms in incorporated references appear to differ from the definitions provided in the present teachings, the definition provided in the present teachings shall control. The section headings used herein are for organizational purposes only and are not to be construed as limiting the described subject matter in any way.

For years it was assumed that *S. coelicolor* produces four compounds: actinorhodin (from the ACT BGC), undecylprodigiosin (from the RED BGC), methylenomycin, and a calcium-dependent antibiotic. Sequence analysis of the genome in 2002 revealed at least 25 pathways for potential secondary metabolites, which led to the discovery that *S. coelicolor* can produce 17 chemically distinct metabolite classes. Whole genome sequencing and computational analysis reveals nearly 1 million BGCs encoding NPs of unknown composition throughout the three domains of life. Decoding the genomes of antibiotic-producing microbes has revealed a surprisingly large number of new pathways, typically ten-fold higher than the number of molecules discovered by traditional approaches. Unfortunately, these pathways are mostly silent; efforts to turn them on have succeeded individually, but not as a large-scale platform. Computational tools to identify interesting pathways (e.g., polyketides, terpenes, nonribosomal peptides, etc.) are readily available, but identifying and finding the associated products is significantly more challenging.

As described herein, the present technology relates to the use of promoters (e.g., inducible promoters) provided in a vector (e.g., an expression vector) flanking a cloning site. In some embodiments, the technology provides a vector (e.g., an expression vector) comprising two promoters that flank a cloning site. After cloning an insert into the cloning site, the promoters flank the insert. Thus, the promoters provided by the expression vector are outside the boundaries of inserts cloned at the cloning site. Further, each promoter of the expression vector faces inward toward the insert and, accordingly, each promoter is upstream of nucleic acid sequences provided by the insert. That is, in some embodiments, the expression vector comprises two promoters (e.g., a first promoter and a second promoter) that direct transcription toward each other and in opposite directions. Accordingly, the two promoters are "face-to-face" promoters or, alternatively, "opposed promoters".

In some embodiments, one or both promoters is/are within 1 to 100 bases of the cloning site (e.g., within 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100

bases of the cloning site). In some embodiments, one or both promoters is/are within 10 to 500 bases of the cloning site (e.g., within 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 290, 295, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350, 355, 360, 365, 370, 375, 380, 385, 390, 395, 400, 405, 410, 415, 420, 425, 430, 435, 440, 445, 450, 455, 460, 465, 470, 475, 480, 485, 490, 495, or 500 bases of the cloning site).

Accordingly, the promoters of the expression vector are capable of transcribing nucleic acids on one or both strands of a cloned insert. According to the technology provided herein, inserts are cloned into expression vectors provided herein without considering reading frame or other relationships between the expression vector promoters and nucleotide sequences of cloned inserts. Thus, the promoters in the expression vector not may or may not be operably linked to one or more genes of the insert. During the development of the technology provided herein, data collected surprising indicated that products were expressed from cloned inserts (e.g., comprising a BGC) without engineered placement of promoters within the insert and in operable linkage with a gene encoded by the insert. The data collected indicated that the promoters (e.g., inducible promoters) provided in the expression vector and outside cloned inserts transcribe nucleic acid of the insert and activate production of gene products (e.g., proteins, biosynthetic pathways comprising proteins, and products produced by biosynthetic pathways and/or proteins) in a heterologous host.

In some embodiments, the technology provides a vector (e.g., an expression vector) comprising a promoter flanking a cloning site, wherein the promoter directs transcription toward the cloning site; and wherein the expression vector is configured to accept a biosynthetic gene cluster nucleic acid at the cloning site and express a product of the biosynthetic gene cluster nucleic acid under control of the promoter.

In some embodiments, the technology provides a vector (e.g., an expression vector) comprising a promoter flanking a cloning site, wherein the promoter directs transcription toward the cloning site; and wherein the expression vector is configured to accept a nucleic acid of at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb) at the cloning site and express a product of the nucleic acid under control of the promoter.

In some embodiments, the technology provides a vector (e.g., an expression vector) comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein said expression vector is configured to accept a biosynthetic gene cluster nucleic acid at the cloning site and express a product of the biosynthetic gene cluster nucleic acid under control of the first promoter and/or the second promoter.

In some embodiments, the technology provides a vector (e.g., an expression vector) comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; and wherein said expression vector is configured to accept a nucleic acid comprising at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb) at the cloning site and

express a product of the nucleic acid under control of the first promoter and/or the second promoter.

In some embodiments, one or both of the promoters is/are provided in multiple (e.g., 2, 3, 4, 5, or more copies), e.g., in a tandem arrangement or with other intervening nucleic acids (e.g., a gene (e.g., an activator and/or repressor gene)). See, e.g., the two PnitA promoters in FIG. 3, in which one promoter transcribes the regulator gene (e.g., NitR) and one promoter transcribes into the insert.

The technology comprises use of promoters that are capable of being introduced into a recombinant nucleic acid construct (e.g., a vector (e.g., an expression vector)) and direct transcription of a cloned insert in a host cell (e.g., a heterologous host cell).

In some embodiments, the present technology comprises use of the Potr or PnitA promoters. In particular, in some embodiments, the technology provides expression vectors, methods of using the expression vectors, and related systems, kits, and uses, wherein the expression vectors comprise a Potr and PnitA promoter flanking a cloning site and the Potr and PnitA promoters direct transcription toward each other and in opposite directions (see, e.g., FIG. 1, FIG. 3, and FIG. 4). As described herein, in some embodiments inserts are cloned into embodiments of the expression vectors provided herein and the technology is used to activate transcription of nucleic acids within the insert (e.g., one or more genes of a BGC and/or an entire BGC and/or a one or more genes of a nucleic acid insert comprising at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb)) to express NP metabolites in a heterologous host.

For example, experiments conducted during the development of the technology indicated successful expression of metabolites from cloned inserts in two *Streptomyces* spp. hosts, *S. lividans* Δred Δact and *S. coelicolor* M1154. The 21-kb ACT cluster and 33-kb RED cluster were cloned from *S. coelicolor* A3 (2) into an embodiment of the dual inducible promoter BAC expression vector ("pDualP"; see, e.g., FIG. 4) in both orientations (see, e.g., FIG. 5). The same inserts comprising the ACT and RED clusters were also cloned into a standard vector without inducible promoters as a control. Both the pDualP constructs and the control constructs were conjugated into *S. lividans* Δred Δact (*S. lividans* comprising deletions of the endogenous nucleic acids (e.g., RED and ACT BGCs) that produce the RED and ACT products). Data collected during experiments described herein indicated that *S. lividans* Δred Δact comprising the control constructs did not produce significant quantities of the red or blue pigments from the RED or ACT native promoters from *S. coelicolor* (see, e.g., FIG. 6, FIG. 7, and FIG. 8). In contrast, expression of the pDualP RED and ACT inducible constructs was clearly activated in *S. lividans* Δred Δact when grown in the presence of one or both inducers of the Potr and PnitA promoters, OTC or ε-cap, respectively (see, e.g., FIG. 8). Wild type *S. lividans* is known to be a poor producer of native ACT or RED pigments (see, e.g., FIG. 8) and the data indicating minimal and/or undetectable expression of ACT or RED pigments by *S. lividans* Δred Δact comprising the control constructs is not unexpected. However, the data indicate the surprising result that inducible promoters placed outside of these cloned heterologous pathways were functionally able to activate both recombinant BGCs in the host cells.

During the development of embodiments of the technology provided herein, data were collected indicating that

novel BGCs discovered from a soil metagenomic library by a next-generation sequencing approach can be cloned into embodiments of the dual-inducible promoter BAC expression vector ("pDualP") to produce increased levels of an antibiotic metabolite relative to the native promoters present within the BGC (see, e.g., FIG. 9). Metagenomic clones P12B21 and P32A16 comprise BGCs that produce metabolites showing relatively weak (e.g., approximately 26%) inhibition of *Acinetobacter baumannii* under control of the unidentified native promoters within the BGC (see, e.g., FIG. 9, black bars). In contrast, the same inserts cloned into the pDualP expression vector and expressed from the pDualP expression vector were detected to produce strong inhibition of *A. baumannii* when activated by the ϵ -cap inducer (FIG. 9, grey bars labeled "B"). In particular, expression of the inserts from the pDualP expression vector produced approximately 59% inhibition of *A. baumannii* by clone P12B21 and approximately 62% inhibition of *A. baumannii* by clone P32A16, which represent a two-fold improvement relative to the control. It is contemplated that optimization of induction time and concentration may reveal even higher levels of inhibition.

Inducible Promoters

As used herein, the terms "Potr" and PnitA" refer to two distinct inducible promoters used for transcribing genes in *Streptomyces* using their cognate inducers oxytetracycline (OTC) and ϵ -caprolactam (ϵ -cap), respectively. See, e.g., Wang et al. (2016) "Development of a Synthetic Oxytetracycline-Inducible Expression System for Streptomyces Using de Novo Characterized Genetic Parts" ACS Synthetic Biology 5:765-73, incorporated herein by reference. The sequences of Potr and PnitA are provided by SEQ ID NOs: 1 and 6, respectively. An engineered derivative of Potr called Potr* is also described in Wang, supra, and is provided by SEQ ID NO: 2.

While the Potr and PnitA promoters are exemplary, the technology is not limited to use of these promoters. Accordingly, the technology includes expression vectors comprising other *Streptomyces* promoters. In some embodiments, the technology comprises use of a constitutive promoter. In some embodiments, the technology comprises use of an inducible promoter. In some embodiments, the technology comprise use of kasOp and its derivatives, synthetic tetracycline-inducible promoter tcp830, the constitutive erythromycin-resistance gene promoter ermEp*, phage 119 promoter SF14p, pstSp and xysAp promoters, thiostrepton-inducible promoter tipAp, synthetic resorcinol-inducible promoter PA3-roIO, actII orf4 promoter, the synthetic cumate-inducible promoter P21-cmt, and/or the 30S ribosomal protein S12 promoter PrpsL. Embodiments provide expression vectors comprising two different promoters flanking a cloning site wherein the two promoters are any two promoters chosen from kasOp and its derivatives, synthetic tetracycline-inducible promoter tcp830, the constitutive erythromycin-resistance gene promoter ermEp*, phage 119 promoter SF14p, pstSp and xysAp promoters, thiostrepton-inducible promoter tipAp, synthetic resorcinol-inducible promoter PA3-roIO, actII orf4 promoter, the synthetic cumate-inducible promoter P21-cmt, and/or the 30S ribosomal protein S12 promoter PrpsL. Further, the technology includes promoters (e.g., constitutive and/or inducible promoters) known in the art for heterologous hosts other than *Streptomyces* spp., e.g., Actinobacteria, Gram-negative hosts (e.g., proteobacterial hosts (e.g., *Pseudomonas* spp., *Agrobacterium* spp.),

Hosts

The technology is not limited in the host organism (e.g., that is transformed with an embodiment of the vector (e.g., an expression vector) provided herein (e.g., a vector comprising an insert (e.g., an insert comprising a BGC and/or an insert comprising at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb), e.g., to express a natural product (e.g., metabolite). The host organism is typically, but not necessarily, a genetically tractable (e.g., culturable under laboratory conditions and manipulable by molecular biological techniques) organism. The host organism may be a member of the domain Bacteria, the domain Eukarya, or the domain Archaea. In some embodiments of the technology, the host microorganism is from the domain Bacteria. In some embodiments, the host organism is a bacterium in the terrabacteria group. In particular embodiments, the host microorganism is from the taxa Actinobacteria, Streptomycetales, or Streptomycetaceae. In some embodiments, the host is from the genus *Streptomyces*. In some embodiments, the host is a *Streptomyces* expression strain, e.g., as defined herein (e.g., *Streptomyces avermitilis*, *Streptomyces venezuelae*, *Streptomyces albus*, *Streptomyces lividans*, and *Streptomyces coelicolor*). In some embodiments, the host organism is a *Streptomyces* spp., e.g., as defined herein.

Sources

Further, the technology is not limited in the source organism, organisms, and/or metagenome from which heterologous nucleic acids (e.g., comprising genes, operons, proteins, pathways, activities, etc.) are obtained for use in cloning as inserts in the expression vectors provided herein. For instance, in some embodiments, the source of the nucleic acid is a member of the domain Bacteria, the domain Eukarya, or the domain Archaea. In some embodiments, the source of the nucleic acid is a cultured Streptomyces. In some embodiments, the source is an organism, plurality of organisms, or metagenomic DNA obtained from the earth (e.g., soil, permafrost, sediments), water (e.g., fresh water, seawater, deep-sea vents), air, materials in the environment (e.g., decaying materials like rotting wood, compost), from the surface (e.g., skin) of animals (e.g., mammals, insects, worms), from inside (e.g., digestive tract, gut) animals (e.g., humans), from plants or plant-associated material (e.g., plant roots, plant seeds), possibly from outer space, and the like. In some embodiments, the source is an organism, plurality of organisms, or metagenomic DNA obtained from man-made or artificial environments (e.g., wastewater, activated sludge, hospitals, and ventilation systems). In general, the source may be procured from natural environments, artificial environments, from attempted replications of natural environments, and the like.

In certain embodiments of the technology, a source nucleic acid that is to be introduced into a host organism may undergo codon optimization to enhance expression of a product. Codon optimization refers to alteration of codons in genes or coding regions of nucleic acids for transformation of an organism to reflect the typical codon usage of the host organism without altering the polypeptide for which the DNA encodes. Codon optimization methods for optimum gene expression in heterologous organisms are known in the art and have been previously described (see, e.g., Welch et al (2009), PLOS One 4: e7002; Gustafsson et al (2004), Trends Biotechnol. 22:346-353; Wu et al (2007), Nucl. Acids Res. 35: D76-79; Villalobos et al (2006), BMC Bioinformatics 7:285; U.S. Pat. App. Pub. No. 2011/

0111413; and U.S. Pat. App. Pub. No. 2008/0292918, each of which is incorporated herein by reference).

Methods

Some embodiments of the technology relate to methods, e.g., methods comprising one or more actions (e.g., steps) described herein. For example, in some embodiments, the technology provides a method of expressing a product from a cloned biosynthetic gene cluster. In some embodiments, methods comprise providing an expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions. Some embodiments of methods comprise a subsequent step of cloning a nucleic acid insert comprising a biosynthetic gene cluster at said cloning site.

In some embodiments, the technology provides a method of expressing a product from a cloned nucleic acid insert comprising at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb). In some embodiments, methods comprise providing an expression vector (e.g., as described herein) comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions. In some embodiments, methods further comprise cloning a nucleic acid insert comprising at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb) at said cloning site. In some embodiments, the nucleic acid insert comprises a biosynthetic gene cluster as described herein.

In some embodiments, methods relate to expressing a product from a cloned biosynthetic gene cluster. For example, in some embodiments, methods comprise providing an expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and second promoter direct transcription toward each other and in opposite directions; cloning a nucleic acid insert comprising a biosynthetic gene cluster at said cloning site to provide a recombinant expression vector comprising said nucleic acid insert; transforming said recombinant expression vector comprising said nucleic acid insert into a host cell; and contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said nucleic acid insert.

In some embodiments, the technology provides a method of expressing a product from a biosynthetic gene cluster. For example, in some embodiments methods comprise providing a host cell comprising a recombinant nucleic acid comprising an expression vector and an insert, wherein said expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and said insert comprises a biosynthetic gene cluster nucleic acid; and contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said biosynthetic gene cluster.

In some embodiments, the technology relates to a method of expressing a product from a nucleic acid insert comprising at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb). For example, in some embodi-

ments, methods comprise providing a host cell comprising a recombinant nucleic acid comprising an expression vector and an insert, wherein said expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and said insert comprises at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb); and contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said nucleic acid insert.

In some embodiments, the technology relates to a method of identifying a nucleic acid comprising a biosynthetic gene cluster. For example, in some embodiments, methods comprise providing a host cell comprising a recombinant nucleic acid comprising an expression vector and an insert, wherein said expression vector comprises a first promoter and a second promoter flanking said insert; the first promoter and second promoter direct transcription toward each other and in opposite directions; and the expression vector is configured to express a product of the insert under control of the first promoter and/or the second promoter; contacting said host cell with an inducer of said first promoter and/or an inducer of said second promoter to induce expression of a product from said insert; detecting expression of said product; and identifying the nucleic acid as a nucleic acid comprising a biosynthetic gene cluster when said product is identified.

In some embodiments, the methods comprise use of a host cell that is a *Streptomyces* spp.

In some embodiments, the nucleic acid insert is from a cultured microorganism. In some embodiments, the nucleic acid insert is from a metagenomic library. In some embodiments, the nucleic acid insert is 5 kb or more, 10 kb or more, or 10, 20, 30, 40, 50, 60, 70, 80, 90, or 100 kb or more (e.g., at least 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb).

In some embodiments, the nucleic acid insert comprises a nucleotide sequence encoding a polyketide synthase (PKS) or a nonribosomal peptide synthase (NRPS). In some embodiments, nucleic acid insert comprises a plurality of genes. In some embodiments, the nucleic acid insert comprises genes encoded by both strands of said nucleic acid insert.

In some embodiments, methods comprise detecting expression of a product encoded by one or more nucleotide sequences of said nucleic acid insert. In some embodiments, methods comprise detecting expression of a product encoded by a biosynthetic gene cluster. In some embodiments, the product is produced by a biosynthetic pathway encoded by nucleic acid and/or the biosynthetic gene cluster. In some embodiments, the product is a biologically active agent. In some embodiments, biologically active agent has antiviral, antimicrobial, antifungal, antiparasitic, or anticancer activity. In some embodiments, the biologically active agent is a polyketide or nonribosomal peptide. In some embodiments, the biologically active agent is a sterol, protein, dye, toxin, enzyme, immunomodulator, immunoglobulin, hormone, neurotransmitter, glycoprotein, radiolabel, radiopaque compound, fluorescent compound, cell receptor protein, cell receptor ligand, antiinflammatory compound, antiglaucomatic agent, mydriatic compound, bronchodilator, local anaesthetic, growth promoting agent, or a regenerative

agent. In some embodiments, the biologically active agent is a terpene, saccharide, or alkaloid. In some embodiments, methods comprise a detecting step that is a selection or a screen.

As used herein, the term “selecting” or “selection” refers to a process of using a selectable marker (e.g., antibiotic resistance gene) and/or selective culturing conditions to select and accordingly obtain host cells that comprise an expression vector and/or nucleic acid insert according to the present disclosure. Successfully transformed host cells can be obtained, e.g., by isolation and/or enrichment from a population of transformed host cells. Successfully transformed host cells are capable of surviving the selection conditions and, in some embodiments, are capable of expressing a product from a cloned insert. Selectable markers and selection systems are widely used to obtain host cells expressing a product of interest, e.g., at a high yield. Respective systems are also useful to generate and identify stably transformed host cells (e.g., clones). One goal of using respective selectable markers and selection systems is to introduce a selectable gene which upon exposure to selective growth conditions allows the identification of cells capable of production of the products of interest. Another goal of using selection systems is to identify a selectable gene present in a cloned insert which upon exposure to selective growth conditions allows the identification of cells capable of production of the products of interest.

As used herein, the term “screen” or “screening” refers to a process of using a screenable marker to identify and accordingly obtain host cells that comprise an expression vector and/or nucleic acid insert according to the present disclosure. Successfully transformed host cells can be obtained, e.g., by observation to detect a signal (e.g., fluorescence or color or some other phenotype) produced by a screenable marker and/or an insert and isolation from a population of transformed host cells. Successfully transformed host cells are capable of producing a detectable signal indicating successful transformation and are capable of expressing a product from a cloned insert. Screenable markers and screening systems are widely used to obtain host cells expressing a product of interest, e.g., at a high yield. Respective systems are also useful to generate and identify stably transformed host cells (e.g., clones). One goal of using respective screenable markers and screening systems is to introduce a gene allows the identification of cells capable of production of the products of interest. Another goal of using screening systems is to identify a gene present in a cloned insert that allows identification of cells capable of production of the products of interest. The terms “selecting” and “screening” apply both to nucleic acids present in the expression vectors as described herein and nucleic acids present in inserts cloned into the expression vectors as described herein.

Systems

In some embodiments, the technology relates to systems for cloning nucleic acid inserts comprising a BGC, nucleic acids encoding a biosynthetic pathway, and/or nucleic acids that are at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb); identifying nucleic acids that comprise a BGC or that encode a biosynthetic pathway; detecting biologically active agents produced by nucleic acids that comprise a BGC, nucleic acids that encode a biosynthetic pathway, and/or nucleic acids that are at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130,

135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb); and/or expressing a product (e.g., a biologically active agent) from nucleic acid inserts comprising a BGC, nucleic acids encoding a biosynthetic pathway, and/or nucleic acids that are at least 10 kb (e.g., at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, or 200 kb). In some embodiments, systems comprise a vector (e.g., an expression vector) as described herein.

In some embodiments, systems further comprise a culture medium. In some embodiments, systems further comprise an inducer of one or more promoters provided on a vector (e.g., an expression vector) provided herein. In some embodiments, systems comprise one or both of the inducers OTC and/or ϵ -cap. In some embodiments, systems comprise a culture dish, tray, plate, or other vessel.

In some embodiments, systems comprise components for automated cell culture and clone management. In some embodiments, systems comprise a computer, e.g., programmed to direct automated cell culture and clone management.

In some embodiments, systems comprise an antibiotic for marker selection. In some embodiments, systems comprise a detector of a signal output by a cell (e.g., a fluorescence detector to detect and/or quantify GFP fluorescence; a colorimeter to detect and/or quantify a colored product; etc.)

Some embodiments of the technology provided herein further comprise functionalities for collecting, storing, and/or analyzing data. For example, in some embodiments the device comprises a processor, a memory, and/or a database for, e.g., storing and executing instructions, analyzing data, performing calculations using the data, transforming the data, and storing the data. Moreover, in some embodiments a processor is configured to control a device or apparatus (e.g., a robot configured to perform one or more actions described herein). In some embodiments, the processor is used to initiate and/or terminate the measurement and data collection. In some embodiments, systems comprise a user interface (e.g., a keyboard, buttons, dials, switches, and the like) for receiving user input that is used by the processor to direct a measurement and/or to control a device or apparatus. In some embodiments, systems further comprise a data output for transmitting data to an external destination, e.g., a computer, a display, a network, and/or an external storage medium.

Uses

The technology finds use in natural products discovery, isolation of nucleic acids encoding BGCs, nucleic acids encoding biosynthetic pathways, and nucleic acids expressing biologically active agents. The technology finds use in metagenomic studies and analysis. The technology finds use in both the commercial and research settings.

Although the disclosure herein refers to certain illustrated embodiments, it is to be understood that these embodiments are presented by way of example and not by way of limitation.

Example 1—Construction of the Dual Inducible Promoter Vector

During the development of embodiments of the technology provided herein, a *Streptomyces* spp. expression construct was synthesized containing dual promoters facing each other with respect to their transcription direction. On one side of the construct are the two elements for oxytetracycline (OTC)-based induction, the OtrR gene and the PotR

promoter adjacent to the cloning site (1). On the other side of the construct are the three elements of ϵ -caprolactam (ϵ -cap)-based induction, the PnitA promoter driving expression of the NitR gene, and a second copy of the PnitA promoter adjacent to the cloning site (2, 3). Both inducible promoters have been validated in multiple *Streptomyces* spp. Between these two promoters is sfGFP (super folder green fluorescent protein) as a control. The dual promoter construct was designed using published information from the individual components (1-3) and then synthesized, and the sequence was verified by ATUM (Newark CA) in an *E. coli* cloning vector. The dual promoter elements were subcloned into the *Streptomyces*-integrative BAC vector pBAC-S in three variants (see FIG. 1). Variant 1 contains the sfGFP under control of OTC. Variant 2 is identical to Variant 1 except that two PacI restriction sites flank the sfGFP. Variant 3 is identical to Variant 2 except that the sfGFP orientation has been flipped to come under control of ϵ -cap. Variant 2 and 3 were tested in *E. coli* with and without inducers and fluorescence indicating apparent expression of sfGFP was observed (see FIG. 2). To make the dual promoter system more useful for subcloning into the majority of BACs conferring chloramphenicol resistance, a version of Variant 2 named pDualP was created with the kanamycin resistance gene in place of the chloramphenicol resistance gene (see FIG. 3).

Example 2—Cloning and Heterologous Expression of Known Colored Antibiotic Genes Red and Act

During the development of embodiments of the technology provided herein, experiments were conducted to test the ability of the dual promoter system to activate clusters upon addition of inducer(s). In particular, the ACT and RED clusters, which encode for actinorhodin (blue pigment) and undecylprodigiosin (red pigment) production, respectively, were captured and cloned directly from *S. coelicolor* A3 (2) genomic DNA. The cells were lysed and genomic DNA (gDNA) extracted and purified. The gDNA was restricted in vitro using Cas9 and two guide RNAs that target sites upstream and downstream of each BGC. Linearized pBAC-S vector was prepared and PCR (polymerase chain reaction) amplification was used to add 40-bp overlaps identical to the ends of the BGC fragment left over after restriction. The linearized vector and the restricted gDNA were incubated together in a DNA assembly (e.g., “Gibson”) reaction from New England Biolabs (Ipswich MA) or Synthetic Genomics (La Jolla CA) that uses the overlaps from the linearized vector and the fragment containing the BGC to produce a circular product. *E. coli* BacOpt 2.0 (Lucigen, Middleton, WI) transformants were screened by colony PCR, restriction digestion pattern of purified plasmid DNA, and Sanger sequencing to confirm cloning of ACT and RED clusters to pBAC-S. To generate dual-promoter versions of RED and ACT, purified plasmid was digested with PacI at sites upstream and downstream of the BGCs. The pDualP vector was restricted with PacI, dephosphorylated, gel purified, and ligated to the PacI-restricted ACT and RED fragments. In addition, two metagenomic BACs containing novel BGCs described below were subcloned in a similar manner using Cas9 restriction (see FIG. 4). After transformation and kanamycin selection, clones were identified by colony PCR for pDualP containing either ACT or RED in each orientation (see FIG. 5). Cultures of *E. coli* BacOpt 2.0 containing each of pDualP, pBAC-S ACT, pBAC-S RED, pDualP ACT (in both orientations), and pDualP RED (in both orientations) were mixed with *E. coli* conjugation

helper strain HB101 (pRK2013) and *S. lividans* Δ act Δ red and plated to MS agar (MS agar contains per liter: 10 g agar, 10 g mannitol, and 10 g soy flour). After 16 hours, the plates were flooded with apramycin to select for transconjugation and nalidixic acid to kill donor and helper *E. coli*. Transconjugants for each construct were isolated and tested on MS agar and R2YE agar (contains per liter: 10 g agar, 104 g sucrose, 0.26 g K₂SO₄, 10.2 g MgCl₂·6H₂O, 10 g glucose, 0.1 g tryptone, 0.25% yeast extract, 0.295 CaCl₂·2H₂O, 0.3% L-proline, 0.573% TES Buffer, 1 mL each trace elements solution, 2.5 mM NaOH) demonstrating inducible expression of pDualP ACT and RED BGCs (see FIG. 6 and FIG. 7). Additionally, YEME or R2YE liquid media were used to further demonstrate inducible expression of the pDualP ACT and RED BGCs (see FIG. 8).

Example 3—Cloning and Identification of Two Novel Metagenomic BGCs

Metagenomic Library Construction. During the development of embodiments of the technology provided herein, experiments were conducted to produce a metagenomic library and identify functional clones in the library. High molecular weight (HMW) metagenomic DNA was isolated from a Cullars Rotation (Auburn, AL, USA) soil plot that had not been amended with fertilizers for the past 100 years. The isolation and purification of soil HMW DNA was conducted by isolating soil microorganisms that were embedded in low melting point agarose, treated with proteinase K, and washed extensively. The agarose was melted and the DNA was sheared by pipetting up to five times to generate DNA in the having a size of approximately 150 kb based on pulsed field gel electrophoresis. The agar was allowed to solidify again, and the DNA was end-repaired with the DNATerminator kit (Lucigen) in a total volume of 500 μ L with 10 μ L of enzymes and then heat killed at 70° C. for 15 minutes. The end-repaired DNA was ligated with BstXI adaptors (10 μ L of 100 μ M each) in a total volume of 700 μ L comprising 10 μ L of ligase (2 U/ μ L, Epicenter), followed by gel fractionation and isolation of large DNA fragments ranging from 100 to 200 kb by pulse-field gel electrophoresis. Purified large DNA fragments (about 100 μ L, 1-3 ng/ μ L) were ligated into the cloning-ready BstXI shuttle vector pSmartBAC-S (16° C. for approximately 18 hours). The ligated DNA mixture was electroporated into competent *E. coli* cells (BAC-Optimized *E. coli* 10G Replicator Cells, Lucigen). Small scale ligations and transformations (1 μ L of DNA per 20 μ L of cells) were used to judge the cloning efficiency. The insert sizes of approximately 50 BAC clones were determined to find conditions that contained the desired insert size. Once the suitability of the trial ligation reaction was confirmed, large-scale ligations and transformations were conducted to achieve 19,200 clones for the BAC library (50×384-well plates arrayed).

Metagenomic Library Sequencing and Identification of novel BGCs. Individual clones from the BAC library were grown in triplicate in 96-well plates using 1 ml LB containing 0.01% arabinose to amplify BAC copy number (4). A three-dimensional pooling strategy was used to combine multiple clones for sequencing in such a way as to enable the location of individual BAC clones. Three pools were made; a row pool, a column pool, and a plate pool. The liquid cultures from each pool were combined as appropriate, the cells were pelleted and the BAC DNA purified as previously described (5). For plates 41-50, the initial pooling strategy merged all 384 clones from each original library plate into a single plate pool (10 plate pools); row clones from the 10

original library plates into single row pools (16 row pools A-P, each pool containing 240 clones); and column clones from the 10 original library plates into single column pools (24 column pools, each pool containing 160 clones). For the remainder of the library (plate no. 1-40), the 384-well plates were replicated in batches of 10 plates into 96 well quadrants. For each batch, 40 plate pools were made from each 96-clone quadrant; 8 row pools A-H were made, one from each 480-clone row (40 quadrant plates $\times$ 12 wells/row); and 12 column pools were made, one from each 320-clone column (40 quadrant plates $\times$ 8 wells/column).

Fragment libraries for sequencing on an Illumina instrument were constructed with 100 ng purified BAC DNA from each pool using the multichannel protocol and reagents from Lucigen (Middleton, WI). Unique indexes were used for each library pool within each batch of 10 library plates (Sets). Libraries were multiplexed and sequenced on Illumina HiSeq 2500 with v3 chemistry at 2 $\times$ 150 bp. The raw HiSeq reads per each column, plate or row pool were imported into the Alabama Super Computer (ASC) to be processed. Reads were filtered for high quality reads (Q score $>$ 30), trimmed, clipped and reads smaller than 30 bp were discarded using the software Trimomatic. To remove host and vector DNA sequences, all processed reads were mapped against *E. coli* DH10B and the vector pBAC-S sequences, and those that did not map to these reference sequences were then assembled using metaSPAdes implementation of SPAdes 3.9.0 software 6. Reads corresponding to each respective sequencing pool were assembled together resulting in 290 sets of contigs.

All contigs generated from SPAdes assembly were tentatively deconvoluted to a clone location using a custom bash script. Briefly, the deconvolution process consisted of renaming each individual contig to include their pool of origin and a unique number identifier. Contigs from the plate pools were compared to those in the column or row pools via BLASTn with 95% identity and a 10 $^{-6}$ e-value cut-off. The BLAST hits were extracted and annotated into 3 categories: 1) completely deconvoluted-plate contigs with hits in both column and row pools; 2) partially deconvoluted-plate contigs with hits in only one other dimension; or 3) singletons-contigs with no significant hits. Once each contig was annotated, the location information in the contig name was used to generate coordinates (plate, column and row) for the respective clone of origin.

A local version of antiSMASH 4.0 with prodigal (meta) for gene prediction was used to predict BGCs from plate pools, which had the greatest coverage per pool. The program was run on a Bioconda environment in the Alabama Supercomputer operating system to afford high-throughput detection. Annotations were conducted by importing the BiosynML antiSMASH 4.0 output into Geneious and manually inspecting BGCs. Selected clones identified as containing an intact BGC were individually grown from the *E. coli* cryostock and the presence of the targeted BGC was confirmed by insert DNA-specific PCR. The isolated BAC DNA was re-sequenced by standard single-end fragment sequencing using a MiSeq sequencer (Illumina, San Diego, CA). Trimming and assembly was conducted with CLC Genomics Workbench 8.5 followed by manual inspection and reassembly was conducted with SPAdes 3.9.0 when necessary. Analysis with antiSMASH 4.0 was conducted as described above for annotation of fully assembled clone insert sequences. Inserts with antiSMASH annotation matching that of their associated contig were considered validated. Clones exhibiting activities of interest were selected for further inspection. Their inserts were fully

annotated using the RAST server (7). RAST and AntiSMASH annotations were combined using Geneious software and were manually inspected. Annotation figures were generated using the package GenoPlotR in R studio (8).

Annotation of Metagenomic inserts of interest (P12B21, P32A16). Inserts of the clones P12B21 and P32A16 were fully annotated in addition to the BGC annotation. Clone P12B21, with an insert of 60,007 bp, has a very short NRPS-like cluster with one complete module; however the “model” sequence prediction spans over 26 kb. The model is followed by efflux ABC-transporter genes possibly linked to antibiotic resistance, and their transcriptional regulator, with a noteworthy presence of a predicted tellurium resistance-linked gene. Clone P32A16 has genes that are most similar to a genomic origin from the phylum Acidobacteria upon RAST annotation. The insert had 59,698 bp and carried 48 features, including a predicted Type I PKS and cell-wall/cell-membrane metabolism genes such as permeases as well as gene sequences predicted to be involved in primary metabolism. The BGC was classified as Type I PKS and encompasses 9 domains distributed in 2 modules, containing condensation domains—suggesting a hybrid NRPS/PKS pathway—as well as a tailoring domain, which may contribute to the structural uniqueness of the compound. Clone P32A16 also contains a predicted TonB-linked transporter and an ABC-ATPase transporter, both with orthologous sequences identified from Acidobacteria taxa, that are in the vicinity of the BGC and may be involved in metabolite secretion.

Example 4—Expression of Antibacterial Activity of Two Metagenomic BGCS from Native or Dual Inducible Promoters

Two BGCs (P12B21 and P32A16) derived from a soil metagenomic library that express an antibacterial metabolite that inhibits the growth of multidrug-resistant *A. baumannii* were subcloned into the pDualP dual-inducible vector and evaluated for inducible expression of antibacterial activity. These pDualP-BGC constructs were transferred by triparental intergeneric conjugation to an expression host (*S. coelicolor* M1154) that was engineered for heterologous expression of BGCs by the removal of four endogenous gene clusters to alleviate precursor competition and the addition of point mutations shown to pleiotropically upregulate antibiotic expression (9). To facilitate the conjugal transfer of each of the BGCs from the donor strain *E. coli* DH10B to the recipient *S. coelicolor* M1154, the helper strain *E. coli* HB101 10 bearing the plasmid pRK2013 11 was used.

Preparation of *E. coli* DH10B donor strains containing a pDual-BGC construct (or pDualP empty vector) for triparental mating was performed by culturing each donor in 2 ml LB liquid medium supplemented with apramycin (50 μ g/ml) at 37 $^{\circ}$ C. overnight. Overnight cultures were then diluted 1:100 in LB containing 50 μ g/ml apramycin and cultured for 4-6 hours until the optical density at 600 nm (OD₆₀₀) reached 0.4 to 0.6. *E. coli* HB101 (pRK2013) was cultured in 1 ml LB supplemented with 30 μ g/ml kanamycin, grown at 37 $^{\circ}$ C. overnight, diluted 1:100 in LB containing kanamycin (30 μ g/ml), and incubated until the OD₆₀₀ was between 0.4 and 0.6. Each *E. coli* donor harboring a separate pDualP-BGC construct and the *E. coli* HB101 (pRK2013) helper strain were pelleted by centrifugation and washed twice in an equal volume of LB to remove antibiotics. *E. coli* donor cells were resuspended in 100 μ l of LB and *E. coli* HB101 (pRK2013) was resuspended in 300 μ l of LB.

Mycelial fragments of *S. coelicolor* M1154 were used as recipients for intergeneric conjugation and were prepared by cultivating *S. coelicolor* M1154 in 20 ml of malt-extract yeast-extract maltose liquid medium (MYM contains per liter: 4 g maltose, 4 g yeast extract, 4 g malt extract) in a flask with a stainless-steel coiled spring, shaking at 200 rpm, 30° C. for 5 days. Mycelia was collected by centrifugation at 3,000×g, washed twice with 2×yeast extract tryptone (2×YT) medium, and resuspended in 400 µl 2×YT medium. Approximately 10⁸ *E. coli* donor cells (100 µl volume of each donor) were mixed with 100 µl of mycelia. The *E. coli*-*S. coelicolor* mixture was pelleted by centrifugation and the pelleted cells were resuspended in the residual liquid after removing most of the supernatant. The mating mixture was spread on mannitol soya flour (MS) agar supplemented with 20 mM MgCl₂ and incubated at 30° C. for 24 hours. The plates were overlaid with 1 ml of sterile water containing 0.5 mg nalidixic acid for counterselection against *E. coli* and 1 mg of apramycin for transconjugant selection. Plates were incubated for an additional 5-7 days at 30° C. until exconjugants were visible, after which exconjugants were replicated to MS plates supplemented with 30 µg/ml nalidixic acid and 50 µg/ml apramycin. Genomic integration of the BGC in each *S. coelicolor* M1154 pDualP exconjugant was validated using PCR analysis.

Screening *S. coelicolor* pDualP clones for inducible expression of antibacterial activity. Quantification of dual-inducible expression of antibacterial activity was performed using a bioassay format in which each metagenomic BGC (n=3) was treated with a single inducer (OTC or ε-cap), both inducers, or no inducers and compared to the expression by the native BGC promoters in *S. coelicolor* M1154. To prepare supernatants for bioassays, *S. coelicolor* pDualP clones were streaked onto MS agar plates and incubated at 30° C. for 4 days. A single colony of each clone was used to inoculate yeast extract-malt extract (YEME) broth and grown at 30° C., shaking at 200 rpm, for 72 hours. Similarly, each of the BGCs cloned in the non-inducible expression system (e.g., native promoter) were cultured in the same manner as the *S. coelicolor* pDualP clones to monitor antibacterial activity with and without promoter-expression capabilities. After 72 hours, *S. coelicolor* pDualP clones were treated with or without 2.5 µM OTC and/or 0.1% (w/v) ε-cap and grown for an additional 96 hours.

Following incubation, mycelium was removed from each *S. coelicolor* culture by centrifugation at 3,000×g for 15 minutes and supernatants were filtered through a 0.2 µm microporous membrane. A volume of 100 µl of cell-free supernatants from each *S. coelicolor* clone with and without the dual-inducible expression system were added to triplicate wells in a 96-well plate. Wells containing supernatants were then mixed with 100 µl of a 1:100 diluted log-phase culture of *A. baumannii* 3806 (12). Additionally, wells containing sterile growth medium (YEME broth containing per liter: 1.5 g yeast extract, 2.5 g Bacto-peptone, 1.5 g malt extract, 5 g glucose, 170 g sucrose, and 2.5 µM MgCl₂) with and without inducers, pathogen with and without inducers, and *S. coelicolor* empty vector treated with and without each inducer were included as negative controls. Plates were incubated for 24 h at 37° C. with shaking at 220 rpm, and the OD₆₀₀ was quantified for each well using a multi-well plate reader. Mean percent inhibition of the pathogen for each clone and treatment was determined relative to the *S. coelicolor* pDualP empty vector negative control. Statistical analyses using pair-wise comparisons derived from linear modeling were conducted in R to evaluate significant differences (at P<0.05) among treatments.

Evaluation of inducible promoter expression and antibacterial activity. Data collected during these experiments indicated a significant increase (>two-fold) in the expression of antibacterial activity when induced with ε-cap (see FIG. 9) for both of the metagenomic BGCs cloned into the pDualP inducible-expression system in comparison to the expression by native promoters. No significant increase in antibacterial activity was observed from OTC induction alone for either of the metagenomic BGCs. Although induction with both ε-cap and OTC increased antibacterial activity for clones P32A16 and P12B21, it is contemplated that this effect was due to the enhanced expression by the ε-cap inducer alone and not by the combination of the two inducers. However, practicing the technology does not require knowledge of the mechanism and is embodiments of the technology are not limited by any particular theory of induction. Thus, induction with ε-cap demonstrated inducible heterologous expression of two metagenomic BGCs in *S. coelicolor* M1154 which is expected to aid in the detection and characterization of the over-produced antimicrobial metabolites.

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All publications and patents mentioned herein, both in this section and throughout the entirety of this application,

are incorporated by reference in their entirety for all purposes. Various modifications and variations of the described compositions, methods, and uses of the technology will be apparent to those skilled in the art without departing from the scope and spirit of the technology as described. Although the technology has been described in connection with specific exemplary embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the described modes for carrying out the invention that are obvious to those skilled in the art are intended to be within the scope of the following claims.

SEQUENCE LISTING

Sequence total quantity: 9

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 mol_type = unassigned DNA
 organism = Streptomyces rimosus

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 organism = Streptomyces rimosus

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 mol_type = unassigned DNA
 organism = Streptomyces rimosus

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 FEATURE Location/Qualifiers
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We claim:

1. A method of expressing a polypeptide from an insert nucleic acid comprising a biosynthetic gene cluster, the method comprising:

providing a nucleic acid comprising a biosynthetic gene cluster;

providing an expression vector comprising a first promoter and a second promoter flanking a cloning site, wherein the first promoter and the second promoter direct transcription toward each other and in opposite directions;

cloning an insert nucleic acid comprising the biosynthetic gene cluster at said cloning site to provide a recombinant expression vector comprising said insert nucleic acid;

transforming a host cell with the recombinant expression vector;

expressing a polypeptide from the biosynthetic gene cluster; and

detecting the expression of the polypeptide using a functional selection or screen that is specific for the polypeptide or for a product that is indicative of the polypeptide.

2. The method of claim 1, wherein said insert nucleic acid is 5 kb or more.

3. The method of claim 1, wherein said insert nucleic acid is 50 kb or more.

4. The method of claim 1, wherein said insert nucleic acid is 100 kb or more.

5. The method of claim 1, wherein said insert nucleic acid comprises an operon with multiple genes.

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6. The method of claim 1, wherein said host cell is a heterologous host.

7. The method of claim 1, further comprising growing the host cell under selective conditions.

8. The method of claim 1, further comprising integrating the recombinant expression vector into the host cell chromosome.

9. The method of claim 1, further comprising contacting the host cell with one or both of an inducer of the first promoter and/or an inducer of the second promoter.

10. The method of claim 1, wherein the polypeptide is a biologically active agent or produces a biologically active agent.

11. The method of claim 1, wherein the insert nucleic acid comprises a nucleotide sequence encoding a polyketide synthase (PKS) or a nonribosomal peptide synthase (NRPS).

12. The method of claim 1, wherein said insert nucleic acid comprises a plurality of genes.

13. The method of claim 1, wherein both strands of said nucleic acid insert comprise genes.

14. The method of claim 1, wherein providing a nucleic acid comprising a biosynthetic gene cluster comprises obtaining a nucleotide sequence of a metagenomic nucleic acid, identifying the biosynthetic gene cluster in the nucleotide sequence, and selecting a clone comprising the biosynthetic gene cluster to provide the insert nucleic acid.

15. The method of claim 1, wherein said product is a metabolite.

16. The method of claim 1, wherein said product has an antimicrobial activity.

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